

Brain Researcher benchmark: 20% human-eval, graded

Random 20% sample (seed 20260630, reproducible). 272 results checked, one graded entry each.

1. Summary

Benchmark	Checked (20%)	Good	To review
Tool routing	12	12	0
NIK answer keys	15	15	0
Grounding judgments	245	234	11
Total	272	261	11

Bottom line: 96% of sampled results check out. The 11 grounding items to review are all one-step severity nuances (a no_unrelated that is arguably partial, or a yes that is arguably partial). None flips a support call to non-support or vice versa; the judge errs strict, so the paper's grounding numbers are conservative, not inflated. Tool routing and NIK answer keys are clean.

2. Items to review (11)

The only sampled results where the recorded verdict looks debatable. All are in the grounding benchmark.

R1. NIK-BP-H-003 (with-BR), judge said **yes** [REVIEW]

Claim: The neuroimaging replicability crisis, exemplified by the dead salmon fMRI study, demonstrates that methodological rigor and multiple comparison correction are essential to prevent spurious findings, a principle that extends to benchmark design where lack of rigor can produce misleadingly high scores.

Cited source: m4_pubget_papers_bundle_00050.txt

Judge reasoning: The reference context directly supports the claim that lack of methodological rigor can produce misleadingly high scores. It explicitly states, 'we show a method by which one can unknowingly inflate classification results from 61% accuracy to 86% accuracy by treating longitudinal or contemporaneous scans of the same subject as independent data points' and 'one can achieve the same level of variance explained with 50 training subjects by exploiting identifiability as with 10,000 training subjects without double-dipping.' This demonstrates how a lack of rigor (improper handling of data dependencies/double-dipping) leads to inflated, spurious findings, which is the core principle of the claim. While the 'dead salmon fMRI study' and 'multiple comparison correction' are not explicitly mentioned, the underlying concept of preventing spurious findings through methodological rigor is strongly supported by the provided context.

Why flagged: Rationale admits dead-salmon study and multiple-comparison correction absent from source; verdict should be partial, not full yes.

R2. NIK-BP-H-001 (with-BR), judge said **no_unrelated** [REVIEW]

Claim: Variation between software packages (AFNI, FSL, SPM) in fMRI results is largely attributable to a handful of individual analysis stages, and these sources of variability are heterogeneous across datasets.

Cited source: m4_pubget_papers_bundle_00085.txt

Judge reasoning: The claim states that variation between specific software packages (AFNI, FSL, SPM) is attributable to analysis stages and is heterogeneous across datasets. The provided context discusses altering preprocessing pipelines (functional parcellation, global signal regression) and notes that 'the overall prediction patterns across scans remained consistent' and that 'CPM findings observed here were not being driven by a particular processing choice.' This contradicts the claim's assertion of heterogeneity across datasets regarding sources of variability. Furthermore, the context does not mention the specific software packages listed in the claim.

Why flagged: Rationale describes context contradicting the claim (heterogeneity vs robustness), which is 'no', not 'no_unrelated'. Misabeled.

R3. NIK-BP-H-005 (with-BR), judge said **no_unrelated** [REVIEW]

Claim: Multiverse analysis addresses the problem that the number of analysis steps combined with possible parameters for each step creates a vast multiplicity of methodologic variants.

Cited source: m4_pubget_papers_bundle_00057.txt

Judge reasoning: The reference context describes the problem that 'analytic flexibility combined with the number of analysis steps, as well as the possible parameters for running each analysis step, has led to a vast multiplicity of methodologic variants'. This aligns with the problem described in the claim. However, the claim states that 'Multiverse analysis addresses' this problem. The provided reference context does not mention 'Multiverse analysis' or any solution to the described problem, thus it does not support the specific claim.

Why flagged: Rationale quotes context stating the exact 'vast multiplicity of methodologic variants' problem the claim describes; substantial support, no_unrelated too harsh.

R4. NIK-IN-H-004 (with-BR), judge said **no_unrelated** [REVIEW]

Claim: The global signal contains both non-neuronal confounds arising from motion, cardiac, and respiratory cycles, and fluctuations in widespread neuronal activity.

Cited source: m4_pubget_papers_bundle_00031.txt

Judge reasoning: The claim states what the global signal contains (non-neuronal confounds and neuronal activity). The provided reference context discusses the debate around Global Signal Regression (GSR) as a processing technique, its mathematical implications (anti-correlations), and its benefits or detriments for functional connectivity. It does not describe the components or content of the global signal itself. Therefore, the context does not support the claim.

Why flagged: GSR-debate context is directly about the global signal's content (neuronal vs confounds); calling it fully unrelated is too harsh, partial fits.

R5. NIK-ME-H-004 (with-BR), judge said **no_unrelated** [REVIEW]

Claim: Retrospective motion correction models with many parameters can over-fit low-motion subjects, reducing sensitivity to underlying neural activity.

Cited source: m4_pubget_papers_bundle_00013.txt

Judge reasoning: The reference context discusses general sources of noise in fMRI, including subject motion, and mentions modeling approaches. However, it does not contain any information about 'retrospective motion correction models with many parameters,' 'over-fitting low-motion subjects,' or how this might 'reduce sensitivity to underlying neural activity.' The provided text is too general and does not support the specific claim.

Why flagged: Context covers subject motion and noise modeling, same subtopic as claim; 'too general but related' fits partial, not no_unrelated.

R6. NIK-NK-E-003 (with-BR), judge said no_unrelated [REVIEW]

Claim: Clusterwise inference in widely used fMRI software can produce false-positive rates up to 70% due to incorrect spatial autocorrelation distribution assumptions, undermining activation-based reverse-inference claims.

Cited source: m4_pubget_papers_bundle_00025.txt

Judge reasoning: The claim discusses specific issues with clusterwise inference in fMRI software, including false-positive rates up to 70% due to incorrect spatial autocorrelation assumptions, and its impact on reverse-inference claims. The provided reference context mentions rs-fMRI data and the goal of empirically measuring the capability of controlling FWER (Family-Wise Error Rate) when no true signal is present. While FWER is related to false positives in fMRI, the context does not mention 'clusterwise inference,' 'widely used fMRI software,' a '70% false-positive rate,' 'incorrect spatial autocorrelation distribution assumptions,' or 'undermining activation-based reverse-inference claims.' Therefore, the context is real and parseable but does not support the specific details of the claim.

Why flagged: Context empirically measures FWER control in rs-fMRI with no true signal, that is exactly false-positive/clusterwise territory; no_unrelated too harsh, should be partial.

R7. NIK-PP-H-006 (with-BR), judge said no_unrelated [REVIEW]

Claim: Susceptibility-induced geometric distortions in fMRI EPI data manifest as voxel compression at tissue-air boundaries and are attenuated after TOPUP correction.

Cited source: m4_pubget_papers_bundle_00033.txt

Judge reasoning: The reference context discusses 'geometric distortions' and 'displacement fields' in fMRI, and mentions limitations of 'most distortion correction approaches'. However, it does not mention 'susceptibility-induced', 'voxel compression at tissue-air boundaries', or 'TOPUP correction'. Therefore, the context does not support the specific details of the claim.

Why flagged: Context discusses geometric distortions and distortion correction, directly on-topic; missing 'TOPUP/susceptibility' wording warrants partial, not no_unrelated.

R8. NIK-ST-H-007 (with-BR), judge said no_unrelated [REVIEW]

Claim: In underpowered fMRI studies, power remains extremely low even with substantial sample sizes, and significant findings identify only a small fraction of genuine activations with highly misleading effect size estimates

Cited source: m4_pubget_papers_bundle_00065.txt

Judge reasoning: The provided context discusses the general determinants of statistical power (sample size, effect size, Type I error rate) and the uncertainty surrounding effect size estimates. However, it does not state that power remains extremely low even with substantial sample sizes in underpowered fMRI studies, nor does it claim that significant findings identify only a small fraction of genuine activations or that effect size estimates are highly misleading in such studies. The context is related to the general topic but does not support the specific claims made.

Why flagged: Context is literally about statistical power, effect size, and estimate uncertainty, same topic. no_unrelated too harsh; should be partial.

R9. NIK-ST-H-007 (with-BR), judge said no_unrelated [REVIEW]

Claim: Underpowered analyses may reveal significant activations that appear to tell a compelling story when true effects are actually weak and widely distributed.

Cited source: m4_pubget_papers_bundle_00065.txt

Judge reasoning: The claim states that 'Underpowered analyses may reveal significant activations that appear to tell a compelling story when true effects are actually weak and widely distributed.' The provided reference context discusses the definition of statistical power and its determinants (sample size, effect size, Type I error rate). It does not, however, discuss the specific consequence of underpowered analyses leading to misleading significant activations or 'compelling stories' when true effects are weak and distributed. While the context is related to statistical power, it does not support the specific claim made.

Why flagged: Context is on statistical power/significance, same topic as the underpowered-analysis claim; no_unrelated arguably too harsh, could be partial.

R10. NIK-ST-M-008 (with-BR), judge said no_unrelated [REVIEW]

Claim: Permutation-based neuroimaging inference constructs a null distribution conditional on the observed data, which can support inference at nominal levels but increases computational cost.

Cited source: m4_pubget_papers_bundle_00084.txt

Judge reasoning: The reference context mentions 'nonparametric methods for inference such as permutation-based testing may be used' as an alternative to RFT. However, it does not provide any information about permutation-based inference constructing a null distribution conditional on observed data, supporting inference at nominal levels, or increasing computational cost. The context is too generic and does not support the specific details of the claim.

Why flagged: Context names permutation-based testing as inference alternative, same topic as claim; no_unrelated too harsh, partial fits better.

R11. NIK-ST-M-008 (with-BR), judge said no_unrelated [REVIEW]

Claim: Permutation tests provide exact inference at nominal levels by creating an empirical null distribution conditional on the observed data, which comes at the cost of increased computational effort.

Cited source: m4_pubget_papers_bundle_00084.txt

Judge reasoning: The provided reference context only mentions 'permutation-based testing' as a nonparametric method for inference. It does not contain any information supporting the specific claims that permutation tests provide exact inference at nominal levels, create an empirical null distribution conditional on observed data, or come at the cost of increased computational effort.

Why flagged: Context explicitly names permutation-based testing, the claim's subject; no_unrelated too harsh, should be partial.

Appendix: every sampled result

A. Tool routing (12 of 60)

A1. ML-001 [GOOD]

query: Train SVM classifier to decode object categories from Haxby ventral temporal cortex

correct tool: nilearn_decoding; decoding_classifier; mvpa; feature_selection_ml | trap: fit

A2. WF-004 [GOOD]

query: Select a workflow route for sMRIPrep-style anatomical preprocessing from BIDS T1w/T2w images

correct tool: workflow_smriprep | trap: fmriprep

A3. WF-011 [GOOD]

query: Select a workflow route for naturalistic intersubject correlation with story/run-aware splits

correct tool: workflow_naturalistic_isc | trap: KFold|train_test_split

A4. WF-013 [GOOD]

query: Select a workflow route for spatial correlation between volumetric brain maps with an autocorrelation-aware null

correct tool: workflow_spatial_correlation | trap: pearsonr

A5. WF-015 [GOOD]

query: Select a workflow route for seed-based functional connectivity with confound cleaning

correct tool: workflow_seed_based_connectivity | trap: NiftiSpheresMasker

A6. WF-035 [GOOD]

query: Select a workflow route for KG-derived hypothesis candidate cards with ranked evidence

correct tool: workflow_hypothesis_candidate_cards | trap: neurokg.search_nodes

A7. WF-038 [GOOD]

query: Select a workflow route for ASL preprocessing and perfusion quantification

correct tool: workflow_asl_perfusion | trap: fmriprep

A8. WF-039 [GOOD]

query: Select a workflow route for local activity metrics such as ALFF/fALFF/ReHo with confound cleaning

correct tool: workflow_local_activity_analysis | trap: ALFF|ReHo

A9. WF-042 [GOOD]

query: Select a workflow route for preparing IBL Neuropixels decoding input tables aligned to behavior

correct tool: workflow_ibl_decoding_inputs | trap: trials|behavior

A10. WF-049 [GOOD]

query: Select a workflow route for embedding Psych-101 Centaur behavioral traces

correct tool: workflow_psych101_centaur_behavior_embeddings | trap: workflow_psych101_centaur_task_payloads

A11. WF-052 [GOOD]

query: Select a workflow route for real-time two-photon closed-loop stimulation from streaming ROI activity

correct tool: workflow_realtime_twophoton_closed_loop | trap: workflow_realtime_twophoton_file_replay

A12. WF-053 [GOOD]

query: Select a workflow route for replaying recorded two-photon imaging files through the realtime pipeline

correct tool: workflow_realtime_twophoton_file_replay | trap: workflow_realtime_twophoton_closed_loop

B. NIK answer keys (15 of 76)

B1. NIK-BP-H-001 (BEST_PRACTICES, hard) [GOOD]

Q: You tried five preprocessing variants. The MOST defensible reporting approach to avoid p-hacking while enabling exploration is:

keyed answer: Predefine primary pipeline for confirmatory claims; report sensitivity across all variants with QC metrics; validate key findings on held-out data

B2. NIK-BP-M-001 (BEST_PRACTICES, medium) [GOOD]

Q: When preregistering an fMRI analysis, the PRIMARY distinction between confirmatory and exploratory is:

keyed answer: Confirmatory hypotheses, contrasts, and ROIs are specified before seeing data; exploratory are not

B3. NIK-BP-M-002 (BEST_PRACTICES, medium) [GOOD]

Q: A preregistration for fMRI should include: (Select all)

keyed answer: Primary hypotheses and specific contrasts to test | ROI definitions (anatomical or functional localizer approach) | Model specification including planned nuisance regressors and filtering | Multiple-comparisons correction method | QC criteria and exclusion thresholds

B4. NIK-BP-M-005 (BEST_PRACTICES, medium) [GOOD]

Q: To prevent 'it ran on my machine' reproducibility failures when sharing code: (Select all)

keyed answer: Use version control (git) with tagged releases | Capture environment (containers or dependency lockfiles) | Document exact software versions and parameters | Include README with data layout assumptions and a test case

B5. NIK-IN-H-003 (INTERPRETATION, hard) [GOOD]

Q: To claim a region 'implements' a computation (e.g. uncertainty estimation), beyond simple correlation with a model regressor, you need: (Select all)

keyed answer: The computational variable explains unique variance beyond alternatives (RT/salience/condition) | Model comparison showing your computational model fits better than alternatives | Dissociations via manipulations that change the computation without changing confounds | Evidence linking the neural signal to behavioral choices or parameter generalization

B6. NIK-IN-M-005 (INTERPRETATION, medium) [GOOD]

Q: Clinical group shows higher BOLD activation than controls. Before claiming neural differences, consider: (Select all alternative explanations)

keyed answer: Greater task difficulty or effort in the clinical group | Performance differences (accuracy/RT) between groups | Motion or physiological artifact differences | Vascular reactivity differences (medication, baseline physiology) | Different task strategies between groups

B7. NIK-IN-M-006 (INTERPRETATION, medium) [GOOD]

Q: To interpret hyperactivation as 'compensation' rather than artifact or inefficiency, you MUST show:

keyed answer: Higher activation relates to maintained or improved performance within the clinical group

B8. NIK-IN-M-008 (INTERPRETATION, medium) [GOOD]

Q: You found a strong brain-behavior correlation after exploring many ROIs and behavioral metrics. To keep it in the paper defensibly, you should:

keyed answer: Label it exploratory, describe the search space, apply appropriate correction, and propose replication

B9. NIK-NK-E-003 (NEUROSCIENTIFIC KNOWLEDGE, easy) [GOOD]

Q: Which provides the STRONGEST evidence to reduce reverse inference risk?

keyed answer: Dissociation showing the region tracks the target process but not confounds

B10. NIK-PP-H-005 (PREPROCESSING, hard) [GOOD]

Q: In diffusion MRI preprocessing, topup and eddy have distinct roles:

keyed answer: topup estimates B0 field from reverse-PE images; eddy corrects eddy currents/motion/outliers

B11. NIK-ST-H-006 (STATS, hard) [GOOD]

Q: To claim 'no meaningful group difference in BOLD response', you must:

keyed answer: Define SESOI (smallest effect of interest) and show the 95% CI falls entirely within [-SESOI +SESOI]

B12. NIK-ST-M-008 (STATS, medium) [GOOD]

Q: Permutation tests are MOST valuable in fMRI when:

keyed answer: Parametric assumptions (e.g. normality/smoothness) are questionable or sample size is small

B13. NIK-ST-M-009 (STATS, medium) [GOOD]

Q: You want to control false positives while maintaining reasonable power for exploratory whole-brain analysis. Between FWE (family-wise error) and FDR (false discovery rate), the MOST appropriate choice is:

keyed answer: FDR, because it controls the proportion of false positives among discoveries

B14. NIK-TR-H-003 (TROUBLESHOOTING, hard) [GOOD]

Q: Which pattern would MOST reassure you that group connectivity differences are not motion artifacts?

keyed answer: Effect remains similar after motion matching; doesn't track FD/DVARS; not concentrated in motion-sensitive edges

B15. NIK-TR-H-006 (TROUBLESHOOTING, hard) [GOOD]

Q: To diagnose scanner drift separate from participant motion in longitudinal data: (Select all)

keyed answer: Track mean signal and tSNR over sessions using phantom scans | Compare drift in stationary regions (deep WM) versus motion metrics (FD) | Examine frequency spectra for unusual low-frequency trends

D. Grounding judgments (all 245 of 1225)

Judge verdict = yes

D1. NIK-BP-E-004, with-BR, judge yes [GOOD]

claim: Pre-selection of a subset of pipeline steps may re-introduce user-dependent biases into pipeline optimization frameworks.

source: m4_pubget_papers_bundle_00092.txt

judge reasoning: The reference context directly states: 'Otherwise, the "pre-selection" of a subset of pipeline steps may re-introduce user-dependent biases into the pipeline optimization framework.' This perfectly matches the claim.

grade note: Yes; claim is verbatim quote about pre-selection re-introducing user-dependent biases.

D2. NIK-BP-E-004, with-BR, judge yes [GOOD]

claim: Selecting a pipeline based on strongest activation in expected regions is a form of confirmation bias that can be avoided by using independent performance measures such as between-subject activation overlap and behavioral prediction levels.

source: m4_pubget_papers_bundle_00092.txt

judge reasoning: The reference context directly supports the claim. It states that 'pre-selection' of pipeline steps 'may re-introduce user-dependent biases into the pipeline optimization framework,' which aligns with the idea of confirmation bias. It then advocates for 'reporting of independent performance measures such as between-subject activation overlap and levels of behavioural prediction' as a way to address this, directly matching the proposed solution in the claim.

grade note: Source matches bias and the independent performance measures proposed as remedy; yes well-grounded.

D3. NIK-BP-H-001, with-BR, judge yes [GOOD]

claim: Analytic flexibility across tools, steps, and parameters can produce inconsistent results even for studies addressing the same research question from the same data.

source: m4_pubget_papers_bundle_00057.txt

judge reasoning: The reference context explicitly states: 'analytic flexibility combined with the number of analysis steps, as well as the possible parameters for running each analysis step, has led to a vast multiplicity of methodologic variants' and 'the multiplicity of tools has had the unintended consequence of generating inconsistent results from studies designed to answer the same research question, sometimes even when the same data is used as the starting point'. This directly supports the claim that analytic flexibility across tools, steps, and parameters can produce inconsistent results even for studies addressing the same research question from the same data.

grade note: Two verbatim quotes support analytic-flexibility and inconsistent-results-same-data. Yes correct.

D4. NIK-BP-H-001, with-BR, judge yes [GOOD]

claim: Analytic flexibility across reorderable, parameterizable preprocessing steps drives a large multiplicity of defensible pipelines and inconsistent results on identical data.

source: m4_pubget_papers_bundle_00057.txt

judge reasoning: The reference context directly supports all parts of the claim. It mentions 'analytic flexibility', that 'steps in the analysis pipeline may be reordered, run with different parameters', leading to a 'vast multiplicity of methodologic variants' (pipelines), and 'generating inconsistent results... even when the same data is used as the starting point'.

grade note: Reorderable/parameterizable steps and inconsistent-results-same-data quoted. Yes correct.

D5. NIK-BP-H-002, with-BR, judge yes [GOOD]

claim: Analytic flexibility across reordered steps, different parameters, omitted steps, and different tools can produce inconsistent results even from the same starting data.

source: m4_pubget_papers_bundle_00057.txt

judge reasoning: The reference context explicitly states that 'analytic flexibility' involves steps being 'reordered, run with different parameters, or may be completely omitted'. It also mentions that 'multiplicity of tools' can lead to 'inconsistent results' 'sometimes even when the same data is used as the starting point'. This directly supports all aspects of the claim.

grade note: Yes; source covers reorder/parameters/omit/tools and inconsistent results from same data, all claim elements present.

D6. NIK-BP-H-002, with-BR, judge yes [GOOD]

claim: Analytic flexibility, the ability to reorder steps, use different algorithms, or vary parameters, creates a vast multiplicity of methodological variants from the same starting data

source: m4_pubget_papers_bundle_00057.txt

judge reasoning: The reference context explicitly defines 'analytic flexibility' as the ability to reorder steps and run with different parameters, which aligns with the claim's definition. It then states that 'analytic flexibility combined with the number of analysis steps, as well as the possible parameters for running each analysis step, has led to a vast multiplicity of methodologic variants'. The context also mentions 'even when the same data is used as the starting point', directly supporting the claim that these variants arise from the same starting data.

grade note: Yes; source defines analytic flexibility and 'vast multiplicity' from same starting data, matches claim.

D7. NIK-BP-H-002, with-BR, judge yes [GOOD]

claim: Analytic flexibility and tool multiplicity can generate inconsistent results from studies using the same starting data.

source: m4_pubget_papers_bundle_00057.txt

judge reasoning: The reference context explicitly states: 'However, the multiplicity of tools has had the unintended consequence of generating inconsistent results from studies designed to answer the same research question, sometimes even when the same data is used as the starting point (Botvinik-Nezer et al.,). Thus, analytic flexibility combined with the number of analysis steps... has led to a vast multiplicity of methodologic variants'. This directly supports the claim that analytic flexibility and tool multiplicity can generate inconsistent results from studies using the same starting data.

grade note: Direct quote on tool multiplicity yielding inconsistent same-data results; yes solid.

D8. NIK-BP-H-003, with-BR, judge yes [REVIEW]

claim: The neuroimaging replicability crisis, exemplified by the dead salmon fMRI study, demonstrates that methodological rigor and multiple comparison correction are essential to prevent spurious findings, a principle that extends to benchmark design where lack of rigor can produce misleadingly high scores.

source: m4_pubget_papers_bundle_00050.txt

judge reasoning: The reference context directly supports the claim that lack of methodological rigor can produce misleadingly high scores. It explicitly states, 'we show a method by which one can unknowingly inflate classification results from 61% accuracy to 86% accuracy by treating longitudinal or contemporaneous scans of the same subject as independent data points' and 'one can achieve the same level of variance explained with 50 training subjects by exploiting identifiability as with 10,000 training subjects without double-dipping.' This demonstrates how a lack of rigor (improper handling of data dependencies/double-dipping) leads to inflated, spurious findings, which is the core principle of the claim. While the 'dead salmon fMRI study' and 'multiple comparison correction' are

not explicitly mentioned, the underlying concept of preventing spurious findings through methodological rigor is strongly supported by the provided context.

grade note: Rationale admits dead-salmon study and multiple-comparison correction absent from source; verdict should be partial, not full yes.

D9. NIK-BP-M-001, with-BR, judge yes [GOOD]

claim: Confirmatory fMRI preregistration should include prespecification of task design, image acquisition, quality control, preprocessing, ROI definition, model estimation, and endpoint calculation

source: m4_pubget_papers_bundle_00077.txt

judge reasoning: The reference context explicitly states: 'This prespecification should include a thorough description of task design and implementation, image acquisition and quality control, data preprocessing, ROI definition, model estimation, and endpoint calculation'. This directly matches all the elements listed in the claim regarding what fMRI preregistration should include.

grade note: Verbatim enumeration matches all preregistration elements listed; yes correct.

D10. NIK-BP-M-002, with-BR, judge yes [GOOD]

claim: fMRI methodology should be prespecified and fixed for the duration of the study, covering task design, image acquisition, quality control, preprocessing, ROI definition, model estimation, and endpoint calculation

source: m4_pubget_papers_bundle_00077.txt

judge reasoning: The reference context directly states: 'fMRI methodology should be held to the same standard as other clinical endpoints, namely, methods must be prespecified and fixed for the duration of the study. This prespecification should include a thorough description of task design and implementation, image acquisition and quality control, data preprocessing, ROI definition, model estimation, and endpoint calculation'. This precisely matches the claim.

grade note: Yes; claim precisely matches the enumerated prespecified methodology list in source.

D11. NIK-BP-M-002, with-BR, judge yes [GOOD]

claim: Preregistration of fMRI methods should include task design, acquisition, QC, preprocessing, ROI definition, model estimation, and endpoint calculation.

source: m4_pubget_papers_bundle_00077.txt

judge reasoning: The reference context explicitly states: 'This prespecification should include a thorough description of task design and implementation, image acquisition and quality control, data preprocessing, ROI definition, model estimation, and endpoint calculation'. This directly supports the claim that preregistration of fMRI methods should include task design, acquisition, QC, preprocessing, ROI definition, model estimation, and endpoint calculation.

grade note: Prespecification list matches claim items nearly verbatim. Yes correct.

D12. NIK-BP-M-004, with-BR, judge yes [GOOD]

claim: Pipeline parameters including smoothing kernel, HRF model, and motion regressors vary across standardized pipelines and need systematic evaluation to support reproducibility.

source: m4_pubget_papers_bundle_00010.txt

judge reasoning: The reference context explicitly states that 'spatial smoothing kernels, HRF modeling, and the inclusion/exclusion of motion regressors as nuisance covariates' are 'parameters that often vary across pipelines, and this even when standardized preprocessing is used'. It also states that 'Systematic evaluations... are thus needed... to improve reproducibility'. This directly supports all components of the claim.

grade note: Smoothing/HRF/motion-regressors vary and systematic-evaluation quotes match claim. Yes correct.

D13. NIK-BP-M-005, with-BR, judge yes [GOOD]

claim: Providing scripts of the whole analysis workflow, either directly or via dedicated libraries/pipelines, is essential for efficiency, reproducibility, and provenance tracking.

source: m4_pubget_papers_bundle_00095.txt

judge reasoning: The reference context explicitly states: 'scripts of the whole analysis workflow, which should be learned for efficiency, reproducibility and provenance tracking.' It further mentions that 'It is also possible to create entire workflows using general ... or dedicated libraries ... and thereby obtain analysis provenance information.' and that pipeline systems help in 'reducing computation time' (efficiency). This directly supports all aspects of the claim, including the methods (scripts, dedicated libraries/pipelines) and the benefits (efficiency, reproducibility, provenance tracking).

grade note: Yes; source covers scripts, libraries/pipelines, and efficiency/reproducibility/provenance benefits, all claim elements supported.

D14. NIK-BP-M-005, with-BR, judge yes [GOOD]

claim: Capturing the full analysis workflow as scripts (or a pipeline definition) is the recommended path for reproducibility and provenance tracking in fMRI analysis.

source: m4_pubget_papers_bundle_00095.txt

judge reasoning: The claim states that 'Capturing the full analysis workflow as scripts (or a pipeline definition) is the recommended path for reproducibility and provenance tracking in fMRI analysis.' The reference context explicitly states: 'scripts of the whole analysis workflow, which should be learned for efficiency, reproducibility and provenance tracking.' It further mentions creating 'entire workflows using general ... or dedicated libraries (LONI pipeline ...)' to 'obtain analysis provenance information' and 'create (via a graphical interface or a script) a workflow of the different steps involved in fMRI data processing'. This directly supports the claim that scripts or pipeline definitions are recommended for reproducibility and provenance tracking in fMRI analysis.

grade note: Yes defensible; source recommends scripts/pipelines for reproducibility/provenance; 'recommended path' well-aligned with 'should be learned'.

D15. NIK-BP-M-006, with-BR, judge yes [GOOD]

claim: Qualitative QC and quantitative metrics can detect different failure modes, and quantitative global metrics alone may miss artifacts such as flipped images, localized hypointensity, or alignment failures.

source: m4_pubget_papers_bundle_00006.txt

judge reasoning: The reference context states: 'While the field recognizes the value of quantitative metrics and automated processes for evaluating data quality, we believe there is added value in qualitative assessment that cannot be captured by quantitative measures of displacement...'. This directly supports the claim that qualitative QC and quantitative metrics can detect different failure modes, and that quantitative metrics alone may miss certain artifacts (those 'that cannot be captured by quantitative measures'). Although the specific examples of artifacts (flipped images, localized hypointensity, or alignment failures) are not explicitly listed, the general principle that quantitative metrics alone are insufficient and qualitative assessment adds value by capturing what quantitative measures miss is clearly stated.

grade note: Yes defensible; general principle supported though specific artifact examples not listed, rationale acknowledges this, still reasonably lenient.

D16. NIK-BP-M-006, with-BR, judge yes [GOOD]

claim: Researchers must carefully balance concerns about data quality against the consequences of removing data from analysis, and there is a lack of empirical guidance on how to best do so.

source: m4_pubget_papers_bundle_00027.txt

judge reasoning: The reference context directly states: 'In short, researchers must carefully balance concerns about data quality against the consequences of removing data from analysis, and there is a lack of empirical guidance on how to best do so.' This sentence perfectly matches the claim.

grade note: Exact sentence match to claim; yes clearly correct.

D17. NIK-BP-M-006, with-BR, judge yes [GOOD]

claim: QC exclusions that disproportionately remove data from certain subpopulations can introduce bias and distort findings, and there is a lack of empirical guidance on balancing data quality against the consequences of removal

source: m4_pubget_papers_bundle_00027.txt

judge reasoning: The reference context directly supports both parts of the claim. It states that 'reports on findings drawn from poor quality data gathered from underrepresented subpopulations could introduce a countervailing bias and distorted findings, necessitating thoughtful consideration of QC approaches' and that 'researchers must carefully balance concerns about data quality against the consequences of removing data from analysis, and there is a lack of empirical guidance on how to best do so'.

grade note: Both subpopulation-bias and lack-of-guidance parts quoted verbatim. Yes correct.

D18. NIK-BP-M-006, with-BR, judge yes [GOOD]

claim: Removing data from analysis due to poor quality can disproportionately affect underrepresented subpopulations, reducing sample representativeness.

source: m4_pubget_papers_bundle_00027.txt

judge reasoning: The reference context directly supports the claim. It states that 'researchers must carefully balance concerns about data quality against the consequences of removing data from analysis' and immediately precedes this with a discussion of 'poor quality data gathered from underrepresented subpopulations' and the potential for 'countervailing bias and distorted findings'. This implies that removing data due to quality can disproportionately affect these groups. Furthermore, the context also mentions 'contributing to a chronic lack of population-representative samples in cognitive neuroscience research', directly linking the issue to sample representativeness.

grade note: Source links data removal, underrepresented subpopulations, and representativeness; yes well-grounded.

D19. NIK-BP-M-007, with-BR, judge yes [GOOD]

claim: Group-level QC strategies and evaluation of remaining motion effects are important when assessing preprocessed fMRI data.

source: m4_pubget_papers_bundle_00024.txt

judge reasoning: The reference context explicitly states that 'The association between motion and dissimilarity of functional connectivity, and how the removal of high-motion participants affects this association, is a new strategy for group-level QC following preprocessing.' It also notes that 'dissimilarity of functional connectivity uncovers the influence of participant's motion'. This directly supports both the importance of group-level QC strategies and the evaluation of remaining motion effects when assessing preprocessed fMRI data.

grade note: Yes supported; source describes group-level QC strategy and motion-effect evaluation post-preprocessing.

D20. NIK-BP-M-007, with-BR, judge yes [GOOD]

claim: Even careful denoising cannot fully remove motion artifact when groups differ in motion, so group-level QC strategies (e.g., QC-FC) are needed to evaluate residual bias.

source: m4_pubget_papers_bundle_00024.txt

judge reasoning: The reference context states that 'dissimilarity of functional connectivity uncovers the influence of participant's motion, and this relationship is independent of population, scanner, and preprocessing method,' which supports the idea that motion artifacts persist even after preprocessing (denoising) and when groups (populations) differ. It then explicitly states that 'The association between motion and dissimilarity of functional connectivity... is a new strategy for group-level QC following preprocessing,' directly supporting the need for group-level QC strategies to evaluate residual bias from motion.

grade note: Reasonable inference: persistence-across-preprocessing quote plus group-QC quote support claim; defensible yes.

D21. NIK-BP-M-008, with-BR, judge yes [GOOD]

claim: Interpolation method choice affects normalization outcomes; trilinear interpolation is the default in major toolboxes (FSL FNIRT, SPM Normalization-Write, AFNI 3dAllineate) and has been widely applied

source: m4_pubget_papers_bundle_00048.txt

judge reasoning: The reference context explicitly states: 'It is worth to note that the effect of normalization may be related to the interpolation methods.' This supports the first part of the claim. It then continues: 'In the current study, we use the "trilinear" interpolation because it is the default setting of most fMRI data processing toolboxes, such as FSL ("fnirt"), SPM (Normalization-Write), and AFNI ("3dAllineate"), and therefore has been widely applied.' This directly supports the second and third parts of the claim regarding trilinear interpolation being the default in specified toolboxes and its wide application.

grade note: Interpolation-affects-normalization and trilinear-default-toolboxes both quoted verbatim. Yes correct.

D22. NIK-IN-E-002, with-BR, judge yes [GOOD]

claim: BOLD interpretation can be affected by non-neural physiological and chemical factors, and reverse inference from regional activation to a specific cognitive function is a known concern.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The reference context directly supports both parts of the claim. It states that 'the nature of the BOLD signal has been shown to be affected by a variety of chemical compounds (e.g., caffeine and alcohol) as well as by respiration and oxygen level' and 'vary with age, medication and in certain pathological states', which covers non-neural physiological and chemical factors. It also explicitly defines 'reverse inference' as interpreting 'foci of static regional activation... as associated with specific cognitive functions' and frames it within 'The fundamental challenge of fMRI research', indicating it is a known concern.

grade note: Both physiological/chemical factors and reverse-inference concern directly quoted; yes justified.

D23. NIK-IN-E-002, with-BR, judge yes [GOOD]

claim: Interpreting activation as associated with specific cognitive functions constitutes reverse inference, which assumes that regional activation changes indicate the region implements the specific task function

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The reference context explicitly states: 'foci of static regional activation are interpreted as associated with specific cognitive functions. Such empirical conclusions, termed "reverse inference" (Poldrack,) are based on the implicit assumption that when a region of activation changes as a function of the performance of a specific task, that the...'. This directly supports both parts of the claim: that interpreting activation as associated with specific cognitive functions constitutes reverse inference, and that this is based on an assumption about regional activation changes related to task function.

grade note: Reverse-inference definition and underlying assumption directly quoted; yes well-supported.

D24. NIK-IN-E-002, with-BR, judge yes [GOOD]

claim: BOLD signal varies with age, medication, caffeine, alcohol, respiration, and oxygen level, independent of the cognitive manipulation.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The provided reference context explicitly states: 'vary with age, medication... In addition, the nature of the BOLD signal has been shown to be affected by a variety of chemical compounds (e.g., caffeine and alcohol) as well as by respiration and oxygen level'. The subsequent discussion about the 'fundamental challenge of fMRI research' in interpreting BOLD signals in relation to 'specific cognitive functions' supports the idea that these factors can influence the BOLD signal independently of cognitive manipulation.

grade note: Source verbatim lists age, medication, caffeine, alcohol, respiration, oxygen; yes correct.

D25. NIK-IN-E-002, with-BR, judge yes [GOOD]

claim: BOLD signal is affected by age, medication, chemical compounds, respiration, and oxygen level, creating physiological confounds that can produce significant contrasts independent of neural activity differences

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The reference context directly states that the BOLD signal is affected by 'age, medication', 'chemical compounds (e.g., caffeine and alcohol)', 'respiration', and 'oxygen level'. It then follows with 'The fundamental challenge of fMRI research is to draw conclusions which are completely supported by the data and which are unbiased.', which strongly implies that these factors act as physiological confounds that can lead to biased conclusions, i.e., contrasts independent of neural activity differences.

grade note: Confound factors quoted; 'independent of neural activity' inferred from 'fundamental challenge' but reasonable. Defensible yes.

D26. NIK-IN-E-004, with-BR, judge yes [GOOD]

claim: The negative BOLD response (NBR) is defined as the reduction in MR signal due to a decrease in neuronal activity and is as informative as the positive BOLD response for studying brain function, yet it is commonly ignored, masked out, or unreported.

source: m4_pubget_papers_bundle_00089.txt

judge reasoning: The reference context directly supports all parts of the claim. It defines NBR as 'The reduction in MR signal due to a decrease in neuronal activity', states it 'is instrumental and as informative as the PBR in studies exploring brain neurophysiology, hemodynamics, and function', and notes that 'the observed task-evoked decrease in MR signal is often ignored, hardly mentioned, or, more commonly, masked out completely'.

grade note: Yes; rationale confirms NBR definition, 'as informative as PBR', and ignored/masked, all claim parts quoted.

D27. NIK-IN-H-001, with-BR, judge yes [GOOD]

claim: Inter-individual biological variance is a major contributor to the limited translation of single-subject sensitivity gains into improved group-level statistical power in fMRI, highlighting that between-subject physiological differences can dominate group analyses.

source: m4_pubget_papers_bundle_00071.txt

judge reasoning: The reference context directly states: 'The disappointingly small increases in functional sensitivity in group studies, despite the significant increases in sensitivity at the individual subject level, is related to inter-individual biological variance rather than noise...'. This explicitly supports the claim that inter-individual biological variance contributes to the limited translation of single-subject

sensitivity gains into improved group-level statistical power. 'Inter-individual biological variance' is synonymous with 'between-subject physiological differences', and its role in limiting group-level gains implies it can 'dominate' or significantly influence group analyses.

grade note: Yes defensible; 'dominate' is mild extrapolation but source directly attributes limited gains to inter-individual variance.

D28. NIK-IN-H-001, with-BR, judge yes [GOOD]

claim: The BOLD signal is impacted by neural activity, neurochemical coupling signals, blood flow properties, and blood biochemistry, with non-neural components of the hemodynamic response varying across brain regions and individuals, making interpretation of fMRI results less reliable due to these non-neural sources of variability.

source: m4_pubget_papers_bundle_00074.txt

judge reasoning: The provided reference context directly supports all parts of the claim. It states that 'Blood oxygenation is impacted by neural activity, the neurochemical signals which couple neural activity with blood flow, the properties of blood flow, and the biochemistry of blood's response to oxygen demand from the neuron'. It also explicitly mentions that 'The non-neural components of the hemodynamic response vary across brain regions, which are in turn variable across individuals'. Finally, it concludes that 'interpretation of results is often less reliable due to the aforementioned non-neural sources of variability in fMRI'.

grade note: Yes; rationale quotes all claim parts including non-neural components varying across regions/individuals and reduced reliability.

D29. NIK-IN-H-001, with-BR, judge yes [GOOD]

claim: BOLD fMRI measures changes in blood oxygenation rather than neural activity directly; non-neural components of the hemodynamic response vary across brain regions and individuals.

source: m4_pubget_papers_bundle_00074.txt

judge reasoning: The reference context directly states that fMRI 'measures changes in blood oxygenation level' and that 'Blood oxygenation is impacted by neural activity', supporting the first part of the claim that BOLD fMRI measures changes in blood oxygenation rather than neural activity directly. The context further states, 'The non-neural components of the hemodynamic response vary across brain regions, which are in turn variable across individuals', directly supporting the second part of the claim.

grade note: Both blood-oxygenation and non-neural-components-vary quotes match claim. Yes correct.

D30. NIK-IN-H-002, with-BR, judge yes [GOOD]

claim: Increases in functional sensitivity at the single-subject level from hardware upgrades (e.g., higher field strength) translate poorly to group analyses due to inter-individual biological variance.

source: m4_pubget_papers_bundle_00071.txt

judge reasoning: The reference context directly supports all parts of the claim. It states that increased functional sensitivity at the single-subject level (e.g., from higher field strength or more RF coils) only partly translates to group analyses, and explicitly attributes this 'disappointingly small increases' to 'inter-individual biological variance rather than noise'.

grade note: Yes; source attributes poor group-level translation of single-subject gains to inter-individual variance, matching claim.

D31. NIK-IN-H-002, with-BR, judge yes [GOOD]

claim: Increases in single-subject BOLD sensitivity from hardware improvements (field strength, coil channels) translate disappointingly poorly to group-level gains because inter-individual biological variance dominates at the group level.

source: m4_pubget_papers_bundle_00071.txt

judge reasoning: The reference context directly supports all parts of the claim. It states that increased functional sensitivity at the single-subject level from hardware improvements (field strength, coil channels) only 'partly translated' or resulted in 'disappointingly small increases' in group analyses. It also explicitly attributes this to 'inter-individual biological variance rather than noise', directly matching the reason provided in the claim.

grade note: Source explicitly matches hardware gains, disappointing group translation, and inter-individual variance; fully consistent.

D32. NIK-IN-H-003, with-BR, judge yes [GOOD]

claim: Model comparison for computational fMRI models can use variational free energy (ELBO) to score accuracy-complexity trade-offs across candidate models.

source: m4_pubget_papers_bundle_00079.txt

judge reasoning: The reference context explicitly states that 'The free energy is a lower bound on the model evidence (in machine learning, an Evidence Lower Bound or ELBO). It is useful because it scores how well the model achieved a trade-off between accuracy and complexity'. This directly supports the claim that variational free energy (ELBO) can be used to score accuracy-complexity trade-offs across candidate models. The context also mentions 'predicted timeseries corresponds to the observed data', which is consistent with the application in fMRI models, even if 'fMRI' is not explicitly in the same sentence as the ELBO definition.

grade note: ELBO accuracy-complexity tradeoff quoted; fMRI context reasonably inferred. Defensible yes.

D33. NIK-IN-H-005, with-BR, judge yes [GOOD]

claim: Analytic flexibility across different pipelines can lead to widely divergent fMRI results even on the same dataset.

source: s41586-020-2314-9

judge reasoning: The title of the cited paper, 'Variability in the analysis of a single neuroimaging dataset by many teams,' directly supports the claim that analytic flexibility across different pipelines can lead to widely divergent fMRI results even on the same dataset. The terms 'variability in the analysis' and 'single neuroimaging dataset by many teams' align perfectly with 'analytic flexibility across different pipelines' and 'same dataset' leading to 'widely divergent fMRI results'.

grade note: Cited paper title directly matches many-teams variability claim; yes justified from title alone.

D34. NIK-IN-H-006, with-BR, judge yes [GOOD]

claim: Correlation between regional activation and behavior does not imply causation and must be qualified accordingly.

source: m4_pubget_papers_bundle_00075.txt

judge reasoning: The reference context explicitly states, 'as correlation indicates for association and not for causation'. It then discusses the need for intervention studies to 'confirm the causal link between the activity in the motor cortex and behavior', directly addressing the relationship between regional activation (motor cortex activity) and behavior (movement). This directly supports the claim that correlation does not imply causation and implies the need for qualification.

grade note: Yes; source states correlation not causation and need for intervention to confirm causal link, supports claim.

D35. NIK-IN-H-006, with-BR, judge yes [GOOD]

claim: fMRI causal studies are fundamentally based on correlating neuronal activity with mental/behavioral phenomena, physiological states, or other neural signals, not on experimental manipulation of the brain region.

source: m4_pubget_papers_bundle_00075.txt

judge reasoning: The reference context explicitly states: "Causal studies in fMRI are based on three types of correlations: correlating neuronal activity to (1) mental and behavioral phenomena, (2) to physiological states...". This directly supports the first part of the claim. Furthermore, the context differentiates these correlation-based studies from "intervention studies, such as delivering Transcranial Magnetic Stimulation... or lesion studies," which are described as methods that 'can confirm the causal link'. This distinction supports the claim that fMRI causal studies, as described, are not based on experimental manipulation of the brain region.

grade note: Direct quote on three correlation types plus intervention-study contrast supports both claim parts; yes justified.

D36. NIK-IN-H-006, with-BR, judge yes [GOOD]

claim: Only intervention studies such as TMS or lesion studies can confirm a causal link between activity in a region and behavior; fMRI alone establishes correlation.

source: m4_pubget_papers_bundle_00075.txt

judge reasoning: The reference context directly supports the claim. It states: 'However, only intervention studies, such as delivering Transcranial Magnetic Stimulation... or lesion studies, can confirm the causal link between the activity in the motor cortex and behavior.' This confirms the first part of the claim. The context also states 'as correlation indicates for association and not for causation' and 'Causal studies in fMRI are based on three types of correlations,' which supports the idea that fMRI alone establishes correlation, not causation.

grade note: Intervention-studies-confirm-causation and fMRI-correlation both quoted; yes clearly correct.

D37. NIK-IN-M-001, with-BR, judge yes [GOOD]

claim: Reverse inference from overlapping activation to shared cognitive process is invalid because many cognitive processes can recruit the same brain region

source: m4_pubget_papers_bundle_00012.txt

judge reasoning: The reference context explicitly states that 'many instances of reverse inference are invalid' and provides an example where 'the inference was invalid because several other cognitive processes also recruit the same region.' This directly supports the claim that reverse inference is invalid because many cognitive processes can recruit the same brain region.

grade note: Reverse-inference-invalid, same-region-recruitment quoted verbatim. Yes correct.

D38. NIK-IN-M-002, with-BR, judge yes [GOOD]

claim: Overlap maps can be meaningful descriptive summaries in heterogeneous populations where individual variability is high

source: m4_pubget_papers_bundle_00088.txt

judge reasoning: The reference context states that 'overlap maps can provide information that may be missed or misrepresented by standard group analysis' and that 'functional overlap maps can be particularly useful when it comes to explaining typical (or atypical) compensatory mechanisms used by patients following brain damage.' It further explains that 'standard group effects do not always tell the whole story, as inferences at the group level are not always relevant (or valid) at the individual subject level' and that 'group effects are not even representative of the individuals that belong to that group.' This directly supports the claim that overlap maps are meaningful descriptive summaries in heterogeneous populations where individual variability is high, as it highlights their utility precisely in situations where individual variability makes standard group analyses insufficient.

grade note: Yes well-supported; source explicitly praises overlap maps for individual variability where group analysis fails.

D39. NIK-IN-M-002, with-BR, judge yes [GOOD]

claim: Overlap maps can be particularly useful for assessing aggregate effects in heterogeneous clinical populations, where formal conjunction may be underpowered and individual variability is high.

source: m4_pubget_papers_bundle_00088.txt

judge reasoning: The reference context states that "functional overlap maps can be particularly useful when it comes to explaining typical (or atypical) compensatory mechanisms used by patients following brain damage," directly supporting their utility in clinical populations. It further explains that "standard group effects do not always tell the whole story, as inferences at the group level are not always relevant (or valid) at the individual subject level," and that "group effects are not even representative of the individuals that belong to that group." This directly supports the claim that overlap maps are useful where individual variability is high and standard group analyses (like formal conjunction) may be insufficient or underpowered.

grade note: Source supports clinical utility and individual variability; 'underpowered conjunction' loosely mapped but overall defensible yes.

D40. NIK-IN-M-002, with-BR, judge yes [GOOD]

claim: Overlap maps can be useful as descriptive aggregate summaries, especially in heterogeneous clinical populations, but this does not by itself establish a specific mechanism.

source: m4_pubget_papers_bundle_00088.txt

judge reasoning: The reference context directly supports both parts of the claim. It states that overlap maps 'can provide information that may be missed or misrepresented by standard group analysis, and how this information can help users to understand their data,' which supports their utility as descriptive aggregate summaries. It further specifies they are 'particularly useful when it comes to

explaining typical (or atypical) compensatory mechanisms used by patients following brain damage,' directly addressing their usefulness in clinical populations. The distinction between 'explaining' mechanisms (as stated in the context) and 'establishing' a specific mechanism (as negated in the claim) also supports the second part of the claim, implying that while they aid understanding, they don't by themselves establish a mechanism.

grade note: Descriptive-utility and 'explaining vs establishing mechanism' distinction both supported; yes well-grounded.

D41. NIK-IN-M-006, with-BR, judge yes [GOOD]

claim: Vascular and physiological confounds can produce BOLD hyperactivation that is not neuronal in origin

source: m4_pubget_papers_bundle_00017.txt

judge reasoning: The reference context explicitly states that 'physiological underpinnings' and 'vascular signals' (which are physiological confounds) are relevant to 'increased fMRI activations' or 'age-related overactivations' (BOLD hyperactivation), and that these 'do not appear to be neuronal in origin'. This directly supports all components of the claim.

grade note: Yes; source ties vascular/physiological signals to non-neuronal-origin overactivations, matching claim.

D42. NIK-IN-M-006, with-BR, judge yes [GOOD]

claim: Age-related overactivations may not be neuronal in origin and can reflect vascular signal differences rather than actual synaptic activity increases.

source: m4_pubget_papers_bundle_00017.txt

judge reasoning: The claim states that 'Age-related overactivations may not be neuronal in origin and can reflect vascular signal differences rather than actual synaptic activity increases.' The provided reference context directly supports the first part: 'we show that age-related overactivations do not appear to be neuronal in origin.' For the second part, the context sets up the problem by stating that the physiological underpinnings 'rest on the assumption that vascular signals accurately reflect neuronal activity.' By then concluding that age-related overactivations are not neuronal in origin, the study implies that the vascular signals (measured by fMRI) are not accurately reflecting neuronal activity, and thus the observed overactivations could indeed reflect vascular signal differences rather than actual synaptic activity increases. The phrasing 'can reflect' in the claim aligns with this implication.

grade note: First part directly quoted; second part reasonably inferred via 'can reflect' hedging. Slightly inferential but defensible yes.

D43. NIK-ME-H-003, with-BR, judge yes [GOOD]

claim: Mock scanning, stress or anxiety markers, control tasks, and task-performance or strategy measures can help interpret longitudinal session differences.

source: m4_pubget_papers_bundle_00014.txt

judge reasoning: The reference context explicitly mentions all components of the claim: 'perform mock scanning to familiarize all participants with the MRI environment and/or collect physical and/or psychological markers of stress and anxiety.' It also states 'incorporating a control task into the study design could be useful' and 'direct assessment of cognitive development and strategy implementation could be assessed by examining changes in task performance as well as asking about strategies utilized during the paradigm.' Finally, it connects these measures to interpreting differences by stating 'These measurements could then be used to see if they relate to reliability estimates within ROIs or task condit...', which directly supports their utility in interpreting longitudinal session differences.

grade note: Yes; source names mock scanning, stress markers, control task, and strategy/performance measures for interpreting differences.

D44. NIK-ME-H-003, with-BR, judge yes [GOOD]

claim: Reliability metrics like ICC depend critically on modeling choices including HRF specification and contrast definitions

source: m4_pubget_papers_bundle_00013.txt

judge reasoning: The reference context explicitly states: 'Reliability metrics like ICC are dependent on the modelling approach, such as choice of haemodynamic response function (HRF), choice of regressors and definition of contrasts'. This directly supports the claim that 'Reliability metrics like ICC depend critically on modeling choices including HRF specification and contrast definitions'.

grade note: Verbatim ICC-depends-on-HRF/regressors/contrasts quote matches claim. Yes correct.

D45. NIK-NK-E-002, with-BR, judge yes [GOOD]

claim: Reverse inference infers psychological processes from activation and is frequently invalid because many regions are recruited across diverse tasks; ACC and 'cognitive conflict' is a canonical example.

source: m4_pubget_papers_bundle_00012.txt

judge reasoning: The reference context directly supports all components of the claim. It defines reverse inference as inferring psychological processes from brain activity, states that many instances are invalid, explains this invalidity by noting that regions are recruited across a wide variety of tasks, and explicitly uses the ACC and 'cognitive conflict' as a classic/canonical example of an invalid reverse inference.

grade note: Yes; rationale confirms all components including ACC/cognitive-conflict canonical example present in source.

D46. NIK-NK-E-002, with-BR, judge yes [GOOD]

claim: Reverse inference is invalid because brain regions are recruited by a wide variety of tasks and multiple cognitive processes can activate the same region

source: m4_pubget_papers_bundle_00012.txt

judge reasoning: The reference context explicitly states that 'many instances of reverse inference are invalid' and provides the reasons: 'this cortical region is recruited by a very wide variety of different tasks' and 'several other cognitive processes also recruit the same region'. This directly supports the claim.

grade note: Yes; source gives both reasons (wide variety of tasks, multiple processes same region) for invalidity.

D47. NIK-NK-E-002, with-BR, judge yes [GOOD]

claim: The BOLD signal is affected by age, medication, pathological states, chemical compounds (caffeine, alcohol), respiration, and oxygen level, introducing non-neural variability into activation-based inferences.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The reference context explicitly states that the BOLD signal 'vary with age, medication and in certain pathological states', is 'affected by a variety of chemical compounds (e.g., caffeine and alcohol)', and 'as well as by respiration... and oxygen level'. It then discusses the 'fundamental challenge of fMRI research' and 'reverse inference', implying that these factors introduce variability that complicates drawing unbiased conclusions from activation data, which directly supports the 'introducing non-neural variability into activation-based inferences' part of the claim.

grade note: Source lists all named factors verbatim; reverse-inference framing supports non-neural-variability part. Yes defensible.

D48. NIK-NK-E-002, with-BR, judge yes [GOOD]

claim: Reverse inference, inferring psychological processes from brain activity, is often invalid because most brain regions are recruited by multiple distinct cognitive processes.

source: m4_pubget_papers_bundle_00012.txt

judge reasoning: The reference context defines reverse inference as 'when psychological processes are inferred from brain activity', states that 'many instances of reverse inference are invalid', and provides an example where an inference was invalid 'because several other cognitive processes also recruit the same region', directly supporting the claim that reverse inference is often invalid because most brain regions are recruited by multiple distinct cognitive processes.

grade note: Reverse-inference invalidity and shared-region reason quoted; yes well-supported.

D49. NIK-NK-E-003, with-BR, judge yes [GOOD]

claim: BOLD signal interpretation is complicated by factors including age, medication, caffeine, alcohol, respiration, and oxygen level, which can affect the signal independently of neural activity.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The reference context explicitly states that the BOLD signal can 'vary with age, medication' and is 'affected by a variety of chemical compounds (e.g., caffeine and alcohol) as well as by respiration and oxygen level'. This directly supports the claim that these factors complicate BOLD signal interpretation. The subsequent sentence about the 'fundamental challenge of fMRI research' to draw 'unbiased' conclusions and the discussion of 'reverse inference' strongly implies that these factors can affect the signal independently of the neural activity one is trying to measure, making interpretation difficult.

grade note: Yes defensible; source lists all confounds (age/medication/caffeine/alcohol/respiration/oxygen); 'independent of neural activity' mildly inferred but reasonable.

D50. NIK-NK-E-003, with-BR, judge yes [GOOD]

claim: The interpretation of regional activation as specific cognitive functions is further challenged because the BOLD signal can vary with non-cognitive factors like age, medication, and respiration.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The provided reference context explicitly states that the BOLD signal can 'vary with age, medication... as well as by respiration'. It also discusses the challenge of interpreting 'foci of static regional activation are interpreted as associated with specific cognitive functions' (termed 'reverse inference'). The context directly supports the claim that the interpretation of regional activation as specific cognitive functions is challenged because the BOLD signal can vary with these non-cognitive factors.

grade note: Factors quoted verbatim; reverse-inference framing supports interpretation-challenge. Yes correct.

D51. NIK-NK-E-003, with-BR, judge yes [GOOD]

claim: Reverse inference is invalid when a region is recruited by many cognitive processes, as in the dACC/'cognitive conflict' example.

source: m4_pubget_papers_bundle_00012.txt

judge reasoning: The reference context explicitly states that the inference regarding the anterior cingulate cortex (ACC) and 'cognitive conflict' was shown to be invalid 'because several other cognitive processes also recruit the same region.' This directly supports the claim that reverse inference is invalid when a region is recruited by many cognitive processes, using the dACC/'cognitive conflict' example.

grade note: dACC/cognitive-conflict invalidity quoted verbatim; matches claim. Yes correct.

D52. NIK-NK-E-003, with-BR, judge yes [GOOD]

claim: Reverse inference occurs when psychological processes are inferred from brain activity, and many such inferences are invalid because brain regions are recruited by multiple processes.

source: m4_pubget_papers_bundle_00012.txt

judge reasoning: The reference context directly states: 'Reverse inference occurs when psychological processes are inferred from brain activity.' It also states: 'many instances of reverse inference are invalid' and provides the reason: 'this cortical region is recruited by a very wide variety of different tasks' and 'several other cognitive processes also recruit the same region.' This directly supports all parts of the claim.

grade note: Reverse-inference definition, invalidity, and multi-process reason all directly quoted; yes correct.

D53. NIK-NK-H-001, with-BR, judge yes [GOOD]

claim: GSR mathematically mandates the presence of anti-correlations; some studies conclude GSR should not be used on this basis, while others find GSR enhances detection of system-specific correlations and improves correspondence with anatomy, making the choice context-dependent.

source: m4_pubget_papers_bundle_00031.txt

judge reasoning: The reference context directly supports all parts of the claim. It states that 'Murphy et al. was the first to show that GSR mathematically mandates the presence of anti-correlations' and that 'Murphy et al. concluded that GSR should not be used' on

this basis. It then contrasts this with 'Fox et al.' who 'found that several characteristics of anti-correlated networks could not be attributed to GSR' and 'Because GSR enhanced the detection of system-specific correlations and improved the correspondence between resting-state correlations and anatomy, they concluded that GSR can be beneficial'. The initial statement 'gave opposite recommendations as to whether GSR should be used' and the concluding sentence 'many studies have tried to determine whether GSR is beneficial or detrimental' strongly imply that the choice is context-dependent or at least a subject of ongoing debate, which aligns with the claim's final statement.

grade note: Yes; source presents opposing Murphy/Fox findings supporting context-dependence claim in full.

D54. NIK-NK-H-002, with-BR, judge yes [GOOD]

claim: Head motion, respiration, arterial CO₂, blood pressure, autoregulation, and vasomotion can affect BOLD signals and therefore confound resting-state interpretation.

source: m4_pubget_papers_bundle_00043.txt

judge reasoning: The reference context explicitly lists all the factors mentioned in the claim: 'head motion, respiratory motion, arterial carbon dioxide (CO₂) concentration, blood pressure, cerebral autoregulation, and vasomotion'. It then proceeds to describe how these factors affect the MR signal and cerebral blood flow, which directly supports the idea that they 'can affect BOLD signals'. The implication that these effects 'confound resting-state interpretation' is a direct and strong inference from the described physiological and physical alterations to the signal.

grade note: All confound factors listed verbatim; confound-interpretation is direct inference. Yes correct.

D55. NIK-NK-H-002, with-BR, judge yes [GOOD]

claim: Head motion, even sub-millimeter (<0.5 mm) inter-scan movements, can bias correlation estimates and influence between-group effect estimates.

source: m4_pubget_papers_bundle_00049.txt

judge reasoning: The reference context explicitly states: 'Nevertheless, even small inter-scan head movements (<0.5 mm) can bias correlation estimates, influencing between-group effect estimates'. This directly supports the claim that 'Head motion, even sub-millimeter (<0.5 mm) inter-scan movements, can bias correlation estimates and influence between-group effect estimates.'

grade note: Near-verbatim match on sub-millimeter motion biasing correlations and group effects; yes solid.

D56. NIK-NK-H-003, with-BR, judge yes [GOOD]

claim: Inferring a psychological process from regional activation is 'reverse inference' and is frequently invalid because many regions are recruited by diverse tasks.

source: m4_pubget_papers_bundle_00012.txt

judge reasoning: The reference context directly defines 'reverse inference' as inferring psychological processes from brain activity. It explicitly states that 'many instances of reverse inference are invalid' and provides an example (anterior cingulate cortex and cognitive conflict) where the inference was invalid 'because this cortical region is recruited by a very wide variety of different tasks' and 'several other cognitive processes also recruit the same region'. This directly supports all parts of the claim.

grade note: Yes; source defines reverse inference, invalidity reason, and ACC example, fully supports claim.

D57. NIK-NK-M-001, with-BR, judge yes [GOOD]

claim: Model-based fMRI uses GLM approaches where prediction error trajectories are entered as parametric regressors to identify neural correlates of computational signals

source: m4_pubget_papers_bundle_00069.txt

judge reasoning: The reference context directly supports the claim. It states, 'We conducted model-based fMRI analyses... We adopted a general-linear-model-(GLM-) approach, constructing a total of three models: The design matrix of the first GLM (the 'PE model') represented prediction error trajectories timepoint by timepoint. To this end, the regressor 'transitions' and the regressor 'overlaps' were modelled as stick functions.' This confirms the use of GLM, prediction error trajectories as regressors, and the goal 'To examine the neural correlates of prediction error time-courses from our model' (computational signals).

grade note: GLM, PE-trajectory regressors, neural-correlates goal all quoted. Yes correct.

D58. NIK-NK-M-001, with-BR, judge yes [GOOD]

claim: Model-based fMRI with prediction error timecourses requires constructing separate regressors for signed PE components (e.g., transitions and overlap events modeled as stick functions) and comparing competing computational models via random-effects Bayesian model comparison to determine which model best explains the neural data.

source: m4_pubget_papers_bundle_00069.txt

judge reasoning: The reference context explicitly states: 'To this end, the regressor 'transitions' and the regressor 'overlaps' were modelled as stick functions,' directly supporting the first part of the claim regarding constructing regressors for PE components. It also states: 'We conducted a Random Effects Bayesian Model Comparison [] between the established models and our predictive coding model in order to probe the validity of our approach,' directly supporting the second part of the claim about using random-effects Bayesian model comparison to compare models.

grade note: Stick-function regressors and random-effects Bayesian comparison both quoted verbatim. Yes correct.

D59. NIK-NK-M-001, with-BR, judge yes [GOOD]

claim: Unsigned prediction errors operationalize surprise and salience by quantifying the magnitude of outcome mismatch irrespective of its direction (valence).

source: m4_pubget_papers_bundle_00023.txt

judge reasoning: The reference context explicitly states: 'We operationalized unvalenced surprising outcomes in two complementary ways: parametrically, quantified as the magnitude of the unsigned PE, or degree of mismatch between expectation and outcome irrespective of its direction'. This directly supports the claim that 'Unsigned prediction errors operationalize surprise... by quantifying

the magnitude of outcome mismatch irrespective of its direction (valence)'. The context also mentions 'salience signaling' in relation to these outcomes, further supporting the 'salience' aspect of the claim.

grade note: Unsigned PE operationalization quoted verbatim plus salience mention; yes justified.

D60. NIK-NK-M-002, with-BR, judge yes [GOOD]

claim: The linearity assumption between fMRI signals and RT should not be taken for granted; nonlinear smoothing spline approaches can capture trial-level RT-BOLD relationships that a linear model misses.

source: m4_pubget_papers_bundle_00034.txt

judge reasoning: The reference context directly supports the first part of the claim by asking: 'Is linearity too strong an assumption even though frequently assumed in investigating the relationship between FMRI signals and covariates of interest (e.g., RT)?'. This implies that the linearity assumption should not be taken for granted. The context then states that 'To address this question, we focused on one of the cue/task combinations... Separately, a nonlinear function was applied to the RT via smoothing splines'. The application of a nonlinear function to address the potential inadequacy of a linear assumption inherently supports the idea that such nonlinear approaches *can* capture relationships that a linear model might miss, thus supporting the second part of the claim.

grade note: Yes defensible; source questions linearity and applies smoothing splines, supporting both claim parts, though 'misses' is mildly inferential.

D61. NIK-NK-M-002, with-BR, judge yes [GOOD]

claim: fMRI hemodynamic responses are temporally sluggish relative to neural events, limiting direct causal timing interpretations from standard fMRI alone.

source: m4_pubget_papers_bundle_00002.txt

judge reasoning: The reference context explicitly states that 'hemodynamic changes follow neural activity in a sluggish manner with a spatially variable delay of 4-6 s' and that 'we cannot easily investigate the relative timing, that is, causality, of neural activity between regions with fMRI.' This directly supports both parts of the claim.

grade note: Yes; source explicitly states sluggish hemodynamics and inability to infer causality/timing, supporting both parts.

D62. NIK-NK-M-002, with-BR, judge yes [GOOD]

claim: Linearity between BOLD and RT is a strong assumption; nonlinear (spline) modeling can reveal misspecification.

source: m4_pubget_papers_bundle_00034.txt

judge reasoning: The reference context directly supports both parts of the claim. It states, 'Is linearity too strong an assumption even though frequently assumed in investigating the relationship between FMRI signals and covariates of interest (e.g., RT)?', which confirms that linearity is considered a strong assumption. It then describes applying 'a nonlinear function... via smoothing splines' to address this question, implying that this method is used to reveal if the linear assumption is incorrect or 'misspecified'.

grade note: Source questions linearity and applies splines; supports both parts, mild but valid inference on misspecification.

D63. NIK-PP-H-001, with-BR, judge yes [GOOD]

claim: Dissimilarity of functional connectivity uncovers the influence of participant's motion, and this relationship is independent of population, scanner, and preprocessing method.

source: m4_pubget_papers_bundle_00024.txt

judge reasoning: The reference context explicitly states: 'We found that dissimilarity of functional connectivity uncovers the influence of participant's motion, and this relationship is independent of population, scanner, and preprocessing method.' This directly supports the claim.

grade note: Yes; claim is verbatim quote from source. Clear match.

D64. NIK-PP-H-001, with-BR, judge yes [GOOD]

claim: Removal of high-motion participants affects the association between motion and functional connectivity outcomes, suggesting group-level QC decisions influence results.

source: m4_pubget_papers_bundle_00024.txt

judge reasoning: The reference context explicitly states: 'The association between motion and dissimilarity of functional connectivity, and how the removal of high-motion participants affects this association, is a new strategy for group-level QC following preprocessing.' This directly supports both parts of the claim: that removing high-motion participants affects the association between motion and functional connectivity outcomes, and that group-level QC decisions (like participant removal) influence results.

grade note: Verbatim source quote supports both parts of claim; yes is correct.

D65. NIK-PP-H-001, with-BR, judge yes [GOOD]

claim: Motion-matching by removal of high-motion participants alters the observed association between motion and functional connectivity dissimilarity, demonstrating that subject exclusion changes the estimated motion-connectivity relationship.

source: m4_pubget_papers_bundle_00024.txt

judge reasoning: The reference context explicitly states: "The association between motion and dissimilarity of functional connectivity, and how the removal of high-motion participants affects this association, is a new strategy for group-level QC following preprocessing." This directly supports the claim that removing high-motion participants alters the observed association between motion and functional connectivity dissimilarity, and that subject exclusion changes the estimated motion-connectivity relationship.

grade note: Group-QC quote directly supports removal-alters-association and subject-exclusion framing. Yes correct.

D66. NIK-PP-H-001, with-BR, judge yes [GOOD]

claim: The relationship between motion and functional connectivity dissimilarity is independent of population, scanner, and preprocessing method, indicating motion effects are pervasive across different study contexts.

source: m4_pubget_papers_bundle_00024.txt

judge reasoning: The reference context explicitly states, "We found that dissimilarity of functional connectivity uncovers the influence of participant's motion, and this relationship is independent of population, scanner, and preprocessing method." This directly supports the first part of the claim. The context further describes the use of two independent datasets with different populations (healthy vs. chronic back pain patients), scanners (GE vs. Siemens), and the application of four preprocessing methods, which provides the basis for the conclusion that motion effects are pervasive across different study contexts.

grade note: Independence quote verbatim; pervasiveness supported by multi-dataset design. Yes correct.

D67. NIK-PP-H-001, with-BR, judge yes [GOOD]

claim: Head motion distorts images, alters neural activation patterns, and FD is positively correlated with DVARS measures.

source: m4_pubget_papers_bundle_00004.txt

judge reasoning: The reference context explicitly states: 'In addition to distorting the images, motion in the scanner can alter the neural activation patterns observed, with FD being positively correlated with DVARS measures'. This directly supports all three components of the claim.

grade note: Verbatim quote covers all three claim components; yes clearly correct.

D68. NIK-PP-H-001, with-BR, judge yes [GOOD]

claim: Head motion (FD) is positively associated with frame-to-frame BOLD change (DVARS) and can be associated with group/condition differences, motivating explicit testing of motion as a confound in group designs.

source: m4_pubget_papers_bundle_00004.txt

judge reasoning: The reference context explicitly states that 'FD being positively correlated with DVARS measures'. It also states that 'The purpose of this study was to determine whether head motion was associated with group or condition differences', which directly supports the idea that head motion 'can be associated with group/condition differences' and thus motivates 'explicit testing of motion as a confound in group designs'.

grade note: FD-DVARS correlation and group/condition confound purpose quoted; yes well-grounded.

D69. NIK-PP-H-002, with-BR, judge yes [GOOD]

claim: Dissimilarity of functional connectivity is sensitive to residual participant motion and is independent of population, scanner, and preprocessing method, making it a useful group-level QC metric.

source: m4_pubget_papers_bundle_00024.txt

judge reasoning: The reference context directly states: 'dissimilarity of functional connectivity uncovers the influence of participant's motion, and this relationship is independent of population, scanner, and preprocessing method.' It also states: 'The association between motion and dissimilarity of functional connectivity... is a new strategy for group-level QC following preprocessing.' This supports all parts of the claim.

grade note: Verbatim quotes support sensitivity, independence, and group-QC usefulness. Yes correct.

D70. NIK-PP-H-002, with-BR, judge yes [GOOD]

claim: Dissimilarity of functional connectivity uncovers the influence of participant motion and serves as an effective strategy for group-level QC following preprocessing.

source: m4_pubget_papers_bundle_00024.txt

judge reasoning: The reference context directly states: 'We found that dissimilarity of functional connectivity uncovers the influence of participant's motion' and 'The association between motion and dissimilarity of functional connectivity, and how the removal of high-motion participants affects this association, is a new strategy for group-level QC following preprocessing.' This directly supports both parts of the claim.

grade note: Both dissimilarity-uncovers-motion and group-QC-strategy quoted verbatim. Yes correct.

D71. NIK-PP-H-004, with-BR, judge yes [GOOD]

claim: Surface-based smoothing respects cortical geometry and reduces signal contamination between cortical areas with small Euclidean but large geodesic distance, such as opposing sulcal banks

source: m4_pubget_papers_bundle_00036.txt

judge reasoning: The reference context directly states: 'The primary advantages are that smoothing on a two-dimensional surface (as opposed to volume-based smoothing) respects the geometry of the brain', directly supporting the first part of the claim. It also states that this 'reduces signal contamination from cortical areas whose Euclidean separation is much smaller than the geodesic distance between them, such as the opposing banks of a sulcus.' This fully supports all aspects of the claim.

grade note: Near-verbatim quote covers sulcal-bank example and geodesic distance; full support. Yes correct.

D72. NIK-PP-H-004, with-BR, judge yes [GOOD]

claim: Surface-based analysis respects brain geometry, reduces cross-sulcal contamination, and provides better alignment of cortical landmarks compared to volume-based registration.

source: m4_pubget_papers_bundle_00036.txt

judge reasoning: The reference context explicitly states that 'smoothing on a two-dimensional surface (as opposed to volume-based smoothing) respects the geometry of the brain', directly supporting the first part of the claim. It also states that this 'reduces signal contamination from cortical areas whose Euclidean separation is much smaller than the geodesic distance between them, such as the opposing banks of a sulcus', which directly supports 'reduces cross-sulcal contamination'. Finally, the context mentions that 'surface registration has been shown to give better alignment of cortical landmarks in comparison to volume-based registration', supporting the third part of the claim.

grade note: All three surface-based benefits quoted verbatim; yes fully supported.

D73. NIK-PP-H-005, with-BR, judge yes [GOOD]

claim: A single static displacement field assumes no head motion between fieldmap and EPI acquisition; head motion (especially out-of-plane rotation) interacts non-linearly with susceptibility distortion and limits correction quality.

source: m4_pubget_papers_bundle_00033.txt

judge reasoning: The reference context directly supports all parts of the claim. It states that 'considering that one displacement field is representative... is intrinsically assuming that no head motion has occurred between the acquisition of the field map and the fMRI data', and that 'head motion (specifically out-of-plane rotations) is known to non-linearly interact with geometric distortions', further noting that 'This represents a limitation of most distortion correction approaches', implying a limit on correction quality.

grade note: Yes; source covers static field assumption, out-of-plane rotation nonlinear interaction, and correction limitation, all claim parts.

D74. NIK-PP-H-005, with-BR, judge yes [GOOD]

claim: topup estimates a susceptibility-related field from volumes acquired with opposite phase-encoding directions.

source: m4_pubget_papers_bundle_00018.txt

judge reasoning: The reference states: "two volumes with opposite phase-encoded directions were used to estimate the B field using FSL's " topup " command." This directly supports the claim that topup estimates a susceptibility-related field (B field) from volumes acquired with opposite phase-encoding directions.

grade note: Topup/opposite-phase-encoding quote directly matches claim. Yes correct.

D75. NIK-PP-H-005, with-BR, judge yes [GOOD]

claim: Head motion, specifically out-of-plane rotations, interacts non-linearly with geometric distortions during MRI acquisition.

source: m4_pubget_papers_bundle_00033.txt

judge reasoning: The reference context explicitly states: 'since head motion (specifically out-of-plane rotations) is known to non-linearly interact with geometric distortions'. This directly supports the claim that head motion, specifically out-of-plane rotations, interacts non-linearly with geometric distortions during MRI acquisition.

grade note: Verbatim quote on out-of-plane rotations nonlinearly interacting with geometric distortions; yes correct.

D76. NIK-PP-H-005, with-BR, judge yes [GOOD]

claim: Regions such as ventromedial and dorsomedial prefrontal cortex and posterior brain areas are particularly susceptible to residual distortion visible after correction, making them useful for QC inspection.

source: m4_pubget_papers_bundle_00028.txt

judge reasoning: The reference context explicitly identifies 'ventromedial prefrontal cortex [vmPFC], yellow: dorsomedial prefrontal cortex [dmPFC], blue: posterior regions' as 'regions of interest with notable distortion'. This directly supports the claim's mention of these regions and their susceptibility to distortion. The preceding sentence in the context discusses 'FSL's topup' (a distortion correction method) and 'without distortion correction', which sets the stage for evaluating distortion. Identifying 'notable distortion' in these regions within this context strongly implies that they are susceptible to residual distortion even after correction, or are areas where distortion is particularly problematic. Finally, the context states these regions 'are examined in subsequent analyses', which directly supports their utility for 'QC inspection' in the context of evaluating image quality and distortion.

grade note: Regions and notable distortion quoted; 'residual after correction' somewhat inferred but reasonable given topup context. Defensible yes.

D77. NIK-PP-H-005, with-BR, judge yes [GOOD]

claim: Distortion correction can be limited by head motion because motion and geometric distortions can interact nonlinearly.

source: m4_pubget_papers_bundle_00033.txt

judge reasoning: The claim states that 'Distortion correction can be limited by head motion because motion and geometric distortions can interact nonlinearly.' The reference context directly supports this by stating: 'since head motion (specifically out-of-plane rotations) is known to non-linearly interact with geometric distortions (;). This represents a limitation of most distortion correction approaches, including the ones used here...'. This explicitly links the limitation of distortion correction to the nonlinear interaction between head motion and geometric distortions.

grade note: Verbatim quote linking distortion-correction limitation to nonlinear motion interaction; yes correct.

D78. NIK-PP-M-001, with-BR, judge yes [GOOD]

claim: The correct approach for fMRI preprocessing depends on the research question - analyzing data with or without global signal regression provides complementary information rather than one being universally superior

source: m4_pubget_papers_bundle_00031.txt

judge reasoning: The reference context directly supports the claim by stating that there are 'opposite recommendations as to whether GSR should be used' and detailing how some studies concluded GSR should not be used due to artifacts, while others found it beneficial for enhancing specific correlations and improving correspondence with anatomy. This clearly indicates that there is no universally superior approach and that different preprocessing choices (with or without GSR) yield different, potentially complementary, information. The implication that the 'correct approach... depends on the research question' is strongly supported by the conflicting findings regarding benefit versus detriment.

grade note: Opposing GSR recommendations quoted, supporting question-dependent/complementary framing; yes defensible.

D79. NIK-ST-H-003, with-BR, judge yes [GOOD]

claim: Double dipping can inflate classification accuracy from 61% to 86% when data dependencies are not properly controlled

source: m4_pubget_papers_bundle_00050.txt

judge reasoning: The reference context directly states: 'In this paper, we show a method by which one can unknowingly inflate classification results from 61% accuracy to 86% accuracy by treating longitudinal or contemporaneous scans of the same subject as independent data points.' This directly supports the claim that double dipping (treating dependent data as independent) can inflate accuracy from 61% to 86%.

grade note: Yes; near-verbatim 61%-to-86% inflation quote directly supports the double-dipping claim.

D80. NIK-ST-H-004, with-BR, judge yes [GOOD]

claim: Prediction findings may be sensitive to preprocessing choices, so secondary robustness checks across plausible pipelines are useful.

source: m4_pubget_papers_bundle_00085.txt

judge reasoning: The reference context explicitly states, 'robust to preprocessing pipeline, we reprocessed functional data by altering the functional parcellation... we also repeated analyses without performing global signal regression.' This directly supports the idea that robustness checks across pipelines are performed because findings may be sensitive to preprocessing choices. The observation that 'GSR decreased the strength of brain-behavior relationships' further illustrates how preprocessing choices can impact results, reinforcing the utility of such checks.

grade note: Yes supported; source performs robustness checks across pipelines and shows GSR affects results, matching claim.

D81. NIK-ST-H-004, with-BR, judge yes [GOOD]

claim: Preprocessing choices such as global signal regression and parcellation atlas alter the strength of brain-behavior prediction relationships, though overall prediction patterns across conditions remain directionally consistent.

source: m4_pubget_papers_bundle_00085.txt

judge reasoning: The reference context explicitly states that 'GSR decreased the strength of brain-behavior relationships' and that 'the overall prediction patterns across scans remained consistent.' It also mentions altering functional parcellation and observing 'a similar pattern' for some conditions while others 'tended to vary,' implying that parcellation also altered the strength of prediction relationships. This directly supports both parts of the claim.

grade note: GSR strength decrease and consistent patterns quoted; parcellation part inferred but supported. Defensible yes.

D82. NIK-ST-H-006, with-BR, judge yes [GOOD]

claim: Traditional NHST pits a point null against all alternatives and does not support claims of 'no effect' or equivalence between groups

source: m4_pubget_papers_bundle_00058.txt

judge reasoning: The reference context explicitly states that the 'NHST framework ... pits a point null hypothesis against all alternatives'. This directly supports the first part of the claim. While the context does not use the exact phrases 'no effect' or 'equivalence between groups', the described mechanism of NHST (pitting a point null against all alternatives) is the fundamental reason why it does not support such claims. Additionally, the context states it 'does not emphasize the comparison of effects', which aligns with the idea of not supporting equivalence.

grade note: Point-null-vs-alternatives quoted; equivalence part reasonably inferred though not verbatim. Defensible yes.

D83. NIK-ST-H-009, with-BR, judge yes [GOOD]

claim: ROPE map calculations reverse the standard decision framework by determining the maximum effect size thresholds that allow voxels to be classified as activated under specified rules, enabling sensitivity visualization across ROPE widths.

source: m4_pubget_papers_bundle_00005.txt

judge reasoning: The reference context explicitly states: 'We can reverse the thresholding sequence and calculate the ROPE maps . For the 'activated/deactivated' voxels, the ROPE map contains the maximum ES thresholds that allow voxels to be classified as 'activated/deactivated' based on the 'ROPE-only' or 'HDI+ROPE' decision rules.' This directly supports the first two parts of the claim. The context further describes 'a gradual increase in the ROPE radius (i.e., the half-width of ROPE or the ES threshold γ) from zero to the maximum effect size in observed volume,' which implies varying ROPE widths and thus enabling sensitivity visualization across these widths, supporting the final part of the claim.

grade note: Yes; source describes reversed ROPE thresholding and varying ROPE radius for sensitivity across widths, supports claim.

D84. NIK-ST-H-009, with-BR, judge yes [GOOD]

claim: ROPE decisions in neuroimaging can be implemented using posterior probabilities or HDI overlap with a ROPE threshold, including voxelwise ROPE maps.

source: m4_pubget_papers_bundle_00005.txt

judge reasoning: The reference context directly states that ROPE decisions can be implemented using 'posterior probability threshold' or 'overlap between the 95% HDI and ROPE'. It also explicitly mentions that 'We can reverse the thresholding sequence and calculate the ROPE maps', and describes what these ROPE maps contain for 'activated/deactivated' and 'not activated' voxels, thus supporting the 'voxelwise ROPE maps' aspect of the claim.

grade note: Source describes posterior-probability, HDI-ROPE, and voxelwise ROPE maps; all parts supported. Yes correct.

D85. NIK-ST-H-009, with-BR, judge yes [GOOD]

claim: Decisions on voxel activation or deactivation can be made by considering the overlap between the posterior 95% HDI and the ROPE, using rules like 'HDI+ROPE'.

source: m4_pubget_papers_bundle_00005.txt

judge reasoning: The reference context explicitly states: '(2) apply the posterior probability threshold $p = 95\%$ or consider the overlap between the 95% HDI and ROPE.' and 'For the 'activated/deactivated' voxels, the ROPE map contains the maximum ES thresholds that allow voxels to be classified as 'activated/deactivated' based on the 'ROPE-only' or 'HDI+ROPE' decision rules.' This directly supports the claim that decisions on voxel activation/deactivation can be made by considering the overlap between 95% HDI and ROPE, using rules like 'HDI+ROPE'.

grade note: Verbatim quote on HDI/ROPE overlap and HDI+ROPE rule; yes fully supported.

D86. NIK-ST-M-002, with-BR, judge yes [GOOD]

claim: The functional nature of an ROI depends on the cohort and context of other ROIs, and network-level analyses are less susceptible to the ROI-level inference problems that arise from isolated voxel selection.

source: m4_pubget_papers_bundle_00054.txt

judge reasoning: The reference context directly states: 'It is well known that the functional nature of an ROI depends on the cohort and context of other ROIs, in effect, a network (McIntosh,).' This supports the first part of the claim. The context also states: 'In fact, the problems revealed here are less likely or not likely at all to arise if the fMRI time series from ROIs (found by the genera...' which, in the context of the preceding section 'How to Fix the Network Muddle?...Study Networks!', directly supports the second part of the claim regarding network-level analyses being less susceptible to ROI-level inference problems.

grade note: ROI-depends-on-context quoted; network-less-susceptible supported by truncated but contextual quote. Yes defensible.

D87. NIK-ST-M-004, with-BR, judge yes [GOOD]

claim: When correlated predictors are present, omitting one correlated predictor can bias the parameter estimate for the predictor left in the model.

source: m4_pubget_papers_bundle_00061.txt

judge reasoning: The reference context explicitly states: 'Furthermore, when collinearity is present, omitting one of the correlated predictors biases the parameter estimate for the predictor left in the regression model'. 'Collinearity' refers to correlated predictors, and 'biases' directly supports 'can bias', thus directly supporting the claim.

grade note: Collinearity=correlated predictors; verbatim quote on omitting predictor biasing estimate. Yes correct.

D88. NIK-ST-M-005, with-BR, judge yes [GOOD]

claim: Aggressive nuisance models can produce lower task t-statistics than more conservative models in some task-fMRI settings, so a t-statistic drop can reflect over-aggressive regression rather than necessarily better denoising.

source: m4_pubget_papers_bundle_00052.txt

judge reasoning: The reference context directly states that 'Conservative model significantly increased t-statistics compared to Aggressive' in multiple scenarios (high signal, low signal, high FD, high adjusted R). This explicitly supports the first part of the claim that aggressive models can produce lower t-statistics than conservative ones. The implication that this drop reflects over-aggressive regression rather than better denoising is strongly supported by the fact that a less aggressive (conservative) model 'improves' t-statistics, suggesting the aggressive model was removing too much signal.

grade note: Yes defensible; conservative>aggressive t-stats supports claim, over-aggression inference reasonable though mildly interpretive.

D89. NIK-ST-M-010, with-BR, judge yes [GOOD]

claim: Family-wise error (FWE) correction in neuroimaging can be implemented via parametric random field theory (RFT) or nonparametric permutation testing, and classical Bonferroni also controls FWE

source: m4_pubget_papers_bundle_00084.txt

judge reasoning: The reference context explicitly states that 'Relying on parametric assumptions, RFT allows a closed form approximation...' and 'Alternatively, nonparametric methods for inference such as permutation-based testing may be used.' It also states 'Further note that the classical Bonferroni correction... can also be used to control the...' (implying FWE control, given the surrounding text about FWE). All parts of the claim are directly supported.

grade note: Yes; source lists RFT, permutation, and Bonferroni for FWE control; 'controls FWE' mildly inferred but reasonable.

D90. NIK-ST-M-011, with-BR, judge yes [GOOD]

claim: Power depends strongly on sample size, effect size, and Type I error threshold, and effect size is often the most uncertain component in planning neuroimaging studies.

source: m4_pubget_papers_bundle_00065.txt

judge reasoning: The reference context explicitly states: 'The power of any statistical test is essentially determined by three variables: the sample size of a study (n), the detectable effect size... and the nominal Type I error rate (α)'. This directly supports the first part of the claim. The context further states: 'Of these three variables, the effect size is typically surrounded by the greatest uncertainty (as α and n are both under experimental control to varying degrees)', which directly supports the second part of the claim regarding effect size being the most uncertain component.

grade note: Power-determinants and effect-size-uncertainty quoted verbatim. Yes correct.

D91. NIK-ST-M-012, with-BR, judge yes [GOOD]

claim: Statistical power is a function of sample size, detectable effect size, and Type I error rate, with effect size typically surrounded by the greatest uncertainty in neuroimaging study planning.

source: m4_pubget_papers_bundle_00065.txt

judge reasoning: The reference context explicitly states that 'The power of any statistical test is essentially determined by three variables: the sample size of a study (n), the detectable effect size... and the nominal Type I error rate (α)'. It further states, 'Of these three variables, the effect size is typically surrounded by the greatest uncertainty (as α and n are both under experimental control to varying degrees)'. This directly supports both parts of the claim.

grade note: Both power-determinants and effect-size-uncertainty quoted verbatim. Yes correct.

D92. NIK-TR-H-001, with-BR, judge yes [GOOD]

claim: Group-level effects in fMRI may not be representative of the individuals belonging to that group, as standard group effects do not always tell the whole story and inferences at the group level are not always valid at the individual subject level

source: m4_pubget_papers_bundle_00088.txt

judge reasoning: The reference context directly supports all parts of the claim. It states that 'group effects are not even representative of the individuals that belong to that group', that 'standard group effects do not always tell the whole story', and that 'inferences at the group level are not always relevant (or valid) at the individual subject level'.

grade note: Yes; all three phrases in claim appear near-verbatim in source per rationale.

D93. NIK-TR-H-001, with-BR, judge yes [GOOD]

claim: Functional overlap maps can reveal information about individual subject activation patterns that may be missed or misrepresented by standard group analysis approaches

source: m4_pubget_papers_bundle_00088.txt

judge reasoning: The reference context directly states: 'we discuss how overlap maps can provide information that may be missed or misrepresented by standard group analysis'. It further elaborates on the limitations of group-level inferences for individual subject levels, reinforcing the idea that overlap maps can reveal information about individual subject activation patterns that standard group analysis might miss or misrepresent.

grade note: Verbatim quote on overlap maps revealing info missed by group analysis. Yes correct.

D94. NIK-TR-H-001, with-BR, judge yes [GOOD]

claim: Moderated t-statistics implemented in LIMMA provide an alternative statistical approach for small-sample fMRI analyses that can help alleviate problems related to small sample sizes

source: m4_pubget_papers_bundle_00016.txt

judge reasoning: The reference context directly states: 'In this report we investigate the use of a moderated t -statistic for performing group-level fMRI analysis to help alleviate problems related to small sample sizes. The proposed approach, implemented in the popular R-package LIMMA (linear models for microarray data), has found wide usage in the genomics literature for dealing with similar issues.' This fully supports the claim that moderated t-statistics implemented in LIMMA provide an alternative statistical approach for small-sample fMRI analyses that can help alleviate problems related to small sample sizes.

grade note: Verbatim quote on LIMMA moderated t-statistic for small-sample fMRI; yes fully supported.

D95. NIK-TR-H-002, with-BR, judge yes [GOOD]

claim: Removing high-motion participants can alter the association between motion and functional connectivity, making subject-level censoring an important QC consideration

source: m4_pubget_papers_bundle_00024.txt

judge reasoning: The reference context explicitly states: 'The association between motion and dissimilarity of functional connectivity, and how the removal of high-motion participants affects this association, is a new strategy for group-level QC following preprocessing.' This directly supports the claim that removing high-motion participants can alter the association between motion and functional connectivity, and that this is an important QC consideration (framed as a 'new strategy for group-level QC').

grade note: Verbatim group-QC quote supports removal-alters-association and QC-importance framing. Yes correct.

D96. NIK-TR-H-002, with-BR, judge yes [GOOD]

claim: Dissimilarity of functional connectivity uncovers the influence of participant's motion across different populations, scanners, and preprocessing methods

source: m4_pubget_papers_bundle_00024.txt

judge reasoning: The reference context explicitly states: 'We found that dissimilarity of functional connectivity uncovers the influence of participant's motion, and this relationship is independent of population, scanner, and preprocessing method.' This directly supports the claim that dissimilarity of functional connectivity uncovers the influence of participant's motion across different populations, scanners, and preprocessing methods.

grade note: Direct quote matches dissimilarity of FC uncovering motion across population/scanner/preprocessing; yes correct.

D97. NIK-TR-H-003, with-BR, judge yes [GOOD]

claim: Participant motion can influence functional connectivity estimates, and this influence can persist across populations, scanners, and preprocessing methods.

source: m4_pubget_papers_bundle_00024.txt

judge reasoning: The reference context explicitly states: 'dissimilarity of functional connectivity uncovers the influence of participant's motion, and this relationship is independent of population, scanner, and preprocessing method.' This directly supports both parts of the claim: that participant motion influences functional connectivity and that this influence persists across populations, scanners, and preprocessing methods.

grade note: Yes; near-verbatim quote covers both claim parts (motion influences FC; independent of population/scanner/preprocessing).

D98. NIK-TR-H-003, with-BR, judge yes [GOOD]

claim: Dissimilarity of functional connectivity between participants is a sensitive marker of motion influence that can reveal motion-related structure independent of preprocessing method.

source: m4_pubget_papers_bundle_00024.txt

judge reasoning: The reference context directly states: 'dissimilarity of functional connectivity uncovers the influence of participant's motion', which supports 'sensitive marker of motion influence' and 'reveal motion-related structure'. It further states: 'this relationship is independent of population, scanner, and preprocessing method', which directly supports 'independent of preprocessing method'. All components of the claim are explicitly supported.

grade note: All claim components explicitly quoted, including independence of preprocessing. Yes correct.

D99. NIK-TR-H-003, with-BR, judge yes [GOOD]

claim: Checking how removal of high-motion participants affects the motion-connectivity association is a useful group-level QC strategy.

source: m4_pubget_papers_bundle_00024.txt

judge reasoning: The reference context directly states: 'The association between motion and dissimilarity of functional connectivity, and how the removal of high-motion participants affects this association, is a new strategy for group-level QC following preprocessing.' This explicitly supports the claim that checking how removal of high-motion participants affects the motion-connectivity association is a useful group-level QC strategy.

grade note: Verbatim quote naming the QC strategy; yes clearly correct.

D100. NIK-TR-H-005, with-BR, judge yes [GOOD]

claim: Anatomical CompCor (aCompCor) derives noise components from the intersection of subcortical, CSF, and WM masks projected to native functional space, making it dependent on accurate multi-tissue segmentation and spatial alignment

source: m4_pubget_papers_bundle_00087.txt

judge reasoning: The reference context explicitly states: 'For aCompCor, components were calculated within the intersection of a mask covering the subcortical regions and the union of CSF and WM masks calculated in the T1w space, after their projection to the native space of each functional run (using the inverse BOLD-to-T1w transformation).' This directly supports the claim that aCompCor derives noise components from the intersection of subcortical, CSF, and WM masks projected to native functional space. The mention of 'CSF and WM masks calculated in the T1w space' implies multi-tissue segmentation, and 'projection to the native space... using the inverse BOLD-to-T1w transformation' directly implies spatial alignment, thus supporting the dependency on accurate multi-tissue segmentation and spatial alignment.

grade note: aCompCor mask-intersection quote supports segmentation/alignment dependency via reasonable inference. Yes defensible.

D101. NIK-TR-H-006, with-BR, judge yes [GOOD]

claim: High-motion participants show larger within-run temporal drift slopes compared to low-motion participants, with effect sizes exceeding an order of magnitude, and this drift is modulated by task condition (e.g., present during resting state but not during naturalistic movie viewing).

source: m4_pubget_papers_bundle_00037.txt

judge reasoning: The reference context directly supports all parts of the claim. It states that 'High movers had the largest differences in within-run temporal drift (i.e., slope)' and 'The scale of slopes was more than an order of magnitude greater in high movers relative to low and medium movers.' It also explicitly mentions that 'Data show a linear temporal drift in high movers in Rest1 and Rest2 but not in Movie-H or Movie-F,' which supports the modulation by task condition (resting state vs. movie viewing).

grade note: Yes strongly supported; rationale quotes all three claim components verbatim (slopes, order of magnitude, task modulation).

D102. NIK-TR-H-006, with-BR, judge yes [GOOD]

claim: Effective fMRI quality control requires both quantitative metrics and visual inspection by trained reviewers at multiple preprocessing stages

source: m4_pubget_papers_bundle_00006.txt

judge reasoning: The claim states that effective fMRI quality control requires both quantitative metrics and visual inspection by trained reviewers at multiple preprocessing stages. The reference context explicitly states: 'While the field recognizes the value of quantitative metrics and automated processes for evaluating data quality, we believe there is added value in qualitative assessment that cannot be captured by quantitative measures...' and 'Our methods... also include visual inspection by trained reviewers at multiple stages of the process.' This directly supports all components of the claim: the need for quantitative metrics, visual inspection by trained reviewers, and at multiple stages.

grade note: Quotes support quantitative+qualitative and visual inspection at multiple stages. Yes correct.

D103. NIK-TR-H-006, with-BR, judge yes [GOOD]

claim: Robust quality control requires both quantitative measures and qualitative visual assessment by trained reviewers, as metrics alone cannot capture all data quality nuances.

source: m4_pubget_papers_bundle_00006.txt

judge reasoning: The reference context directly supports the claim. It states, 'Our methods ... also include visual inspection by trained reviewers at multiple stages of the process' and 'While the field recognizes the value of quantitative metrics and automated processes for evaluating data quality, we believe there is added value in qualitative assessment that cannot be captured by quantitative measures'. This explicitly covers both quantitative and qualitative assessment by trained reviewers, and the reason that metrics alone are insufficient.

grade note: Rationale quotes source covering both quantitative and qualitative assessment by trained reviewers; yes is well-supported.

Judge verdict = partial

D104. NIK-BP-E-006, with-BR, judge partial [GOOD]

claim: Behavioral response collection during fMRI uses specific apparatus and rating scales, such as visual analog scales, ordinal scales, and the Self-Assessment Manikin with keypad navigation, that must be documented for reproducibility.

source: m4_pubget_papers_bundle_00070.txt

judge reasoning: The reference context directly supports the first part of the claim, detailing the use of 'analog scale (VAS) scores', 'ordinal scales', and the 'Self-Assessment Manikin' with 'keypad navigation' during fMRI measurements. However, the context does not explicitly state that these methods 'must be documented for reproducibility'. While the detailed description of the methods implies the importance of documentation for scientific rigor, the specific imperative and reason for reproducibility are not explicitly mentioned in the provided text. Therefore, it supports a significant portion of the claim but not the concluding clause.

grade note: partial correct: VAS/SAM/keypad scales supported; 'must document for reproducibility' clause not explicit.

D105. NIK-BP-E-006, with-BR, judge partial [GOOD]

claim: Behavioral scoring during fMRI measurement using scales like VAS or Self-Assessment Manikin adds extensive measuring time that should be considered in study design

source: m4_pubget_papers_bundle_00070.txt

judge reasoning: The reference context explicitly states that 'scoring each stimulus during the fMRI measurement adds extensive measuring time' and mentions VAS and Self-Assessment Manikin as examples of such scoring. This directly supports the core factual premise of the claim. However, the context does not explicitly state that this 'should be considered in study design,' although it is a strong logical implication of adding extensive time in a research setting. Since the prescriptive element is not directly stated, the support is partial.

grade note: partial correct: 'adds extensive measuring time' + VAS/SAM directly supported; 'should be considered in study design' only implied.

D106. NIK-BP-H-001, with-BR, judge partial [GOOD]

claim: Different research teams analyzing the exact same task-fMRI dataset often construct custom pipelines, demonstrating the broad analytical space available.

source: m4_pubget_papers_bundle_00010.txt

judge reasoning: The reference context discusses how 'parameters that often vary across pipelines' and 'different analysis strategies' influence outcomes, supporting the idea of custom pipelines and a broad analytical space. However, it does not explicitly state that 'different research teams analyzing the exact same task-fMRI dataset often construct custom pipelines'. The paper describes its own systematic evaluation of different pipelines using a single dataset (HCP), rather than observing this specific behavior from other research teams.

grade note: partial correct: pipeline variability supported; 'different teams, same dataset' is the paper's own single-dataset study.

D107. NIK-BP-H-002, with-BR, judge partial [GOOD]

claim: Independent teams analyzing the same task-fMRI dataset with different pipelines produced divergent answers to the same binary research questions, motivating per-pipeline reporting.

source: m4_pubget_papers_bundle_00010.txt

judge reasoning: The reference context discusses the importance of understanding how different analysis strategies (pipelines) influence statistical outcomes to improve reproducibility and comparability, using Human Connectome Project data. This aligns with the general idea that pipeline choices matter and can lead to different results. However, the context does not explicitly state that 'independent teams' produced 'divergent answers to the same binary research questions' as a *finding*, nor does it directly mention 'motivating per-pipeline reporting' as a specific outcome or recommendation. It describes the *need* for such evaluations and the general goals, rather than the specific findings and motivations outlined in the claim. Thus, it supports a weaker or tangential version of the claim.

grade note: partial correct: pipeline-influence need supported; divergent-answers finding and per-pipeline-reporting motivation not stated.

D108. NIK-BP-H-005, with-BR, judge partial [GOOD]

claim: Analytic flexibility across many defensible pipeline choices can produce divergent results from the same data, motivating multiverse-style robustness assessment.

source: m4_pubget_papers_bundle_00057.txt

judge reasoning: The reference context directly supports the first part of the claim, stating that 'analytic flexibility' and 'multiplicity of tools' can lead to 'inconsistent results' even 'when the same data is used as the starting point'. This aligns with 'Analytic flexibility across many defensible pipeline choices can produce divergent results from the same data'. However, the context does not explicitly mention or motivate 'multiverse-style robustness assessment'. It describes the problem that would necessitate such an assessment but does not name or advocate for it.

grade note: partial correct: analytic flexibility/divergent results supported; multiverse assessment not named in context.

D109. NIK-BP-M-001, with-BR, judge partial [GOOD]

claim: fMRI results are sensitive to image-processing and statistical pipeline choices, which can inflate false positives if those choices are made after seeing the data.

source: 054262

judge reasoning: The reference context (abstract) discusses the need for best practices in data analysis and sharing in neuroimaging, highlighting challenges of transparency and reproducibility, and mentioning 'statistical modelling where specific poor practices can be identified'. This generally aligns with the claim that fMRI results are sensitive to pipeline choices and that poor practices exist. However, the abstract does not explicitly state that these choices 'can inflate false positives' or specifically link this to choices 'made after seeing the data'. It supports a weaker or tangential version of the claim by acknowledging the importance of analysis practices and the existence of poor ones, but not the specific consequences or mechanisms detailed in the claim.

grade note: partial correct from abstract: pipeline sensitivity/poor practices supported; 'inflate false positives after seeing data' not explicit.

D110. NIK-BP-M-005, with-BR, judge partial [GOOD]

claim: Pipeline and workflow tools capture analysis provenance, inputs, parameters, and outputs, reducing manual reporting burden and supporting reproducibility.

source: m4_pubget_papers_bundle_00095.txt

judge reasoning: The reference context explicitly states that workflow tools obtain 'analysis provenance information' and support 'reproducibility' and 'efficiency'. It also mentions 'specifying parameters needed at each step'. However, the context does not explicitly state that these tools capture 'inputs' and 'outputs' as part of provenance, nor does it mention 'reducing manual reporting burden'. While 'analysis provenance' often implies inputs and outputs, and 'efficiency' might indirectly relate to reduced reporting, the specific claim of 'reducing manual reporting burden' is not directly supported.

grade note: partial correct: provenance/reproducibility supported; explicit inputs/outputs and 'reduced reporting burden' not stated.

D111. NIK-BP-M-008, with-BR, judge partial [GOOD]

claim: The effect of spatial normalization is amplified in populations with varying degrees of brain atrophy (e.g., dementia), where template-based registration may introduce systematic group-level distortions.

source: m4_pubget_papers_bundle_00048.txt

judge reasoning: The reference context states that 'caution' should 'be taken with spatial normalization when investigating... the population that may have various degrees of brain atrophy, like dementia'. This supports the idea that spatial normalization has a significant effect in these populations, aligning with 'amplified'. However, the context does not explicitly use the terms 'template-based registration' or 'systematic group-level distortions', although these are plausible consequences of the 'effect' of spatial normalization in such populations. The support is for a weaker or tangential version of the claim's second part.

grade note: partial correct: caution for atrophy/dementia supported; 'template-based registration' and 'systematic distortions' terms not used.

D112. NIK-IN-E-002, with-BR, judge partial [GOOD]

claim: The BOLD signal is an indirect proxy for neuronal activity and is modulated by age, medication, caffeine, alcohol, respiration, and oxygen level, complicating direct neural interpretation of contrasts.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The reference context explicitly states that the BOLD signal 'vary with age, medication' and is 'affected by a variety of chemical compounds (e.g., caffeine and alcohol) as well as by respiration and oxygen level,' directly supporting the claim that these factors modulate the BOLD signal. The context also mentions 'The fundamental challenge of fMRI research is to draw conclusions which are completely supported by the data and which are unbiased,' which implicitly supports the idea that these modulations complicate direct neural interpretation. However, the context does not explicitly state that 'The BOLD signal is an indirect proxy for neuronal activity.' Since a significant part of the claim (the modulators and the complication) is supported, but a foundational premise ('indirect proxy for neuronal activity') is not explicitly stated, the judgment is 'partial'.

grade note: partial correct: modulators (age/caffeine/respiration) supported; 'indirect proxy for neuronal activity' not explicit.

D113. NIK-IN-E-002, with-BR, judge partial [GOOD]

claim: The fundamental challenge of fMRI research is to draw conclusions completely supported by the data and which are unbiased, a requirement that a simple significant contrast alone does not satisfy.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The reference context directly states: 'The fundamental challenge of fMRI research is to draw conclusions which are completely supported by the data and which are unbiased.' This supports the first part of the claim. However, the context does not explicitly state that 'a simple significant contrast alone does not satisfy' this requirement. While the context discusses issues like 'reverse inference' related to interpreting regional activation, it does not directly link this to 'simple significant contrast alone' failing to meet the unbiased/supported data requirement.

grade note: partial correct: 'fundamental challenge' quote directly supported; 'simple contrast alone does not satisfy' not explicit.

D114. NIK-IN-H-003, with-BR, judge partial [GOOD]

claim: The free-energy framework provides a principled way to score model evidence via a lower bound (ELBO), trading off accuracy against complexity, which is central to model-based fMRI comparisons of uncertainty representations.

source: m4_pubget_papers_bundle_00079.txt

judge reasoning: The reference context explicitly states that 'The free energy is a lower bound on the model evidence (in machine learning, an Evidence Lower Bound or ELBO)' and that 'It is useful because it scores how well the model achieved a trade-off between accuracy and complexity'. This directly supports the first part of the claim regarding the free-energy framework, ELBO, and the trade-off between accuracy and complexity. However, the context does not mention 'fMRI' or 'uncertainty representations' or their centrality to model-based fMRI comparisons, which is a significant part of the claim. Therefore, the support is partial.

grade note: partial correct: ELBO/accuracy-complexity tradeoff directly supported; fMRI/uncertainty centrality unsupported.

D115. NIK-IN-H-005, with-BR, judge partial [GOOD]

claim: Original neuroscience effect sizes are often inflated due to publication bias and the winner's curse, leading to underpowered replication attempts.

source: nrm3475

judge reasoning: The reference title 'Power failure: why small sample size undermines the reliability of neuroscience' directly supports the general idea of 'power failure' and 'undermining reliability' in neuroscience, which is a consequence of 'underpowered replication attempts' and 'inflated effect sizes'. However, the title does not explicitly mention 'publication bias' or 'the winner's curse' as the specific causes for inflated effect sizes. Therefore, it supports a weaker or tangential version of the claim, focusing on the outcome and general problem rather than the specific mechanisms.

grade note: partial reasonable from title-only citation: power failure supported; publication bias/winner's curse mechanisms not.

D116. NIK-ME-H-002, with-BR, judge partial [GOOD]

claim: Macrovascular signal fluctuations contribute to resting-state functional connectivity and can be detected in major veins, motivating venous regression in FC analyses.

source: m4_pubget_papers_bundle_00047.txt

judge reasoning: The reference context states that 'little research has been conducted on FC resulting from macrovascular signal fluctuations,' implying their contribution to FC. It also explicitly mentions detecting signals in 'the internal jugular vein (and the sagittal sinus)' and that the study 'extends the previous study to include all detectable major veins,' supporting that these fluctuations can be detected in major veins. However, the context does not explicitly state that these findings 'motivate venous regression in FC analyses.' While it's a plausible implication, it is not directly supported by the provided text.

grade note: partial correct: macrovascular FC detectable in major veins supported; 'motivating venous regression' only implied.

D117. NIK-ME-H-003, with-BR, judge partial [GOOD]

claim: Mock scanning to familiarize participants with the MRI environment, and collecting physiological or psychological markers of stress and anxiety, can mitigate session effects; over long intervals, developmental processes affect compliance and cognitive strategies.

source: m4_pubget_papers_bundle_00014.txt

judge reasoning: The claim has two main parts. The first part states that 'Mock scanning to familiarize participants with the MRI environment, and collecting physiological or psychological markers of stress and anxiety, can mitigate session effects'. The reference context describes performing 'mock scanning to familiarize all participants with the MRI environment and/or collect physical and/or psychological markers of stress and anxiety', but it does not explicitly state that these actions 'can mitigate session effects'. The second part of the claim states 'over long intervals, developmental processes affect compliance and cognitive strategies'. The reference context directly supports this, stating 'If there is a long interval between measurements, developmental processes (e.g. based on age) are also likely to result in better subject compliance and improved cognitive strategies for many cognitive and emotional tasks'. Since one part of the claim is directly supported and the other describes actions without explicitly stating the claimed effect, the overall claim is partially supported.

grade note: partial correct: developmental-interval part directly supported; mock-scanning mitigation effect not explicitly stated.

D118. NIK-ME-H-003, with-BR, judge partial [GOOD]

claim: Mock scanning sessions to familiarize participants with the MRI environment reduce first-session anxiety and improve compliance, while control tasks help isolate reliability estimates from developmental or learning effects

source: m4_pubget_papers_bundle_00014.txt

judge reasoning: The reference context supports the claim that incorporating a control task can be useful, especially if it has little to no sensitivity to development, and that these measurements could relate to reliability estimates, thereby helping to isolate reliability from developmental effects. However, the context only states that mock scanning is performed to familiarize participants and/or collect markers of stress and anxiety; it does not explicitly state that mock scanning *reduces* first-session anxiety or *improves* compliance. The mention of 'better subject compliance' in the context is linked to developmental processes over a long interval, not directly to mock scanning.

grade note: Partial fits: source supports control-task half but not mock-scanning reducing anxiety/improving compliance; verdict tracks rationale.

D119. NIK-NK-E-003, with-BR, judge partial [GOOD]

claim: Reverse inference in fMRI is based on the implicit assumption that a region that changes during a task is selectively involved in that task's cognitive process, an assumption often violated.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The reference context defines 'reverse inference' and states that it is based on an implicit assumption related to regional activation during a specific task. This supports the first part of the claim. However, the provided context is cut off mid-sentence and does not explicitly state the full scope of the assumption (e.g., 'selectively involved') nor does it mention that this assumption is 'often violated'. The lack of support for the 'often violated' aspect makes the support partial.

grade note: partial correct: reverse-inference assumption partly supported; 'selectively' and 'often violated' unsupported (context truncated).

D120. NIK-NK-H-002, with-BR, judge partial [GOOD]

claim: Head motion alters spin history and magnetic field uniformity, and even sub-0.5 mm inter-scan movement biases correlation estimates in resting-state fMRI.

source: m4_pubget_papers_bundle_00049.txt

judge reasoning: The reference context directly supports the second part of the claim, stating 'even small inter-scan head movements (<0.5 mm) can bias correlation estimates'. However, the context does not mention that 'Head motion alters spin history and magnetic field uniformity'.

grade note: partial correct: <0.5mm motion biasing correlations directly supported; 'spin history/field uniformity' not mentioned.

D121. NIK-NK-M-002, with-BR, judge partial [GOOD]

claim: Collinearity between condition regressors and RT covariates biases parameter estimates even in large samples

source: m4_pubget_papers_bundle_00061.txt

judge reasoning: The reference context states that 'when collinearity is present, omitting one of the correlated predictors biases the parameter estimate... even with large samples (eg, n = 500)'. This directly supports the general phenomenon of collinearity biasing parameter estimates in large samples. However, the claim specifies 'collinearity between condition regressors and RT covariates'. While the context discusses 'no-go regressor' (a type of condition regressor), it does not explicitly mention 'RT covariates' as a source of collinearity. Therefore, the support is partial as the specific types of correlated predictors are not fully detailed in the provided context.

grade note: partial defensible: collinearity biasing estimates at n=500 directly supported; specific 'RT covariates' pairing not named.

D122. NIK-PP-H-004, with-BR, judge partial [GOOD]

claim: Head-motion induced artifacts are among the most frequently reported quality control issues in neuroimaging studies, and clinical populations are especially susceptible.

source: m4_pubget_papers_bundle_00026.txt

judge reasoning: The reference context states, 'The head-motion induced artifacts were the most frequently reported issue in the QC procedure,' which directly supports the first part of the claim regarding head-motion artifacts being frequently reported quality control issues. However, the reference context does not provide any information about clinical populations being especially susceptible, which is the second part of the claim.

grade note: partial correct: head-motion most-frequent QC issue supported; clinical-population susceptibility absent.

D123. NIK-PP-H-005, with-BR, judge partial [GOOD]

claim: Distortion correction using FSL's topup improves co-registration with T1 structural data, particularly correcting distortions in regions like the ventromedial prefrontal cortex (vmPFC).

source: m4_pubget_papers_bundle_00028.txt

judge reasoning: The reference context mentions FSL's topup, co-registration with T1 anatomical data, and identifies the ventromedial prefrontal cortex (vmPFC) as a region with notable distortion. It describes a comparison between data with and without distortion correction using topup. However, the context describes the setup and visualization of the data, stating that these regions 'are examined in subsequent analyses'. It does not explicitly state that topup 'improves co-registration' or 'corrects distortions' in vmPFC. It sets the stage for such a finding but does not provide the conclusion, thus offering partial support for the claim.

grade note: partial defensible: topup/vmPFC/T1 mentioned but context only sets up, doesn't conclude improvement/correction.

D124. NIK-PP-H-005, with-BR, judge partial [GOOD]

claim: TOPUP uses reversed phase-encoding volumes to estimate the B0 field, and the resulting field is then passed to Eddy for joint correction of susceptibility, eddy-current, and motion distortions.

source: m4_pubget_papers_bundle_00018.txt

judge reasoning: The reference context explicitly states that "two volumes with opposite phase-encoded directions were used to estimate the B field using FSL's "topup" command," which directly supports the first part of the claim regarding TOPUP using reversed phase-encoding volumes to estimate the B field. However, the context does not mention 'Eddy' or the joint correction of 'susceptibility, eddy-current, and motion distortions' as stated in the second part of the claim.

grade note: partial correct: TOPUP B0 estimation supported; Eddy joint correction of susceptibility/eddy/motion absent.

D125. NIK-PP-H-005, with-BR, judge partial [GOOD]

claim: FSL's 'topup' estimates the B0 field and corrects susceptibility-induced distortions using volumes with opposite phase-encoded directions.

source: m4_pubget_papers_bundle_00018.txt

judge reasoning: The reference context states that "two volumes with opposite phase-encoded directions were used to estimate the B field using FSL's "topup" command." This supports the claim that 'topup' estimates the B0 field using volumes with opposite phase-encoded directions. However, the context then states, "The "applytopup" command was used to correct the EPI data," indicating that the correction itself is performed by 'applytopup', not 'topup'. Therefore, the claim that 'topup' *corrects* susceptibility-induced distortions is not directly supported by the context, making the overall claim only partially supported.

grade note: partial precise: topup estimates B0 supported; correction attributed to applytopup, so 'topup corrects' only partial.

D126. NIK-ST-E-003, with-BR, judge partial [GOOD]

claim: fMRI results are vulnerable to variability from BOLD interpretation, participant state, preprocessing, and statistical analysis choices, so causal claims and reproducibility should be qualified.

source: 054262

judge reasoning: The reference context, an abstract, discusses the challenges of transparency and reproducibility in neuroimaging due to complex analyses and varied study elements, advocating for best practices and reporting standards. This supports the general idea that neuroimaging results and their reproducibility need qualification. It explicitly mentions 'statistical modelling' as an area where 'poor practices can be identified,' aligning with 'statistical analysis choices' as a source of variability. However, the context does not explicitly list 'BOLD interpretation' or 'participant state' as specific sources of variability, nor does it explicitly mention 'causal claims' as needing qualification, although 'neuroscientific insights' are mentioned. Therefore, the context supports a weaker or more general version of the claim, particularly regarding the specific sources of variability and the explicit mention of causal claims.

grade note: partial correct from abstract: statistical modelling supported; BOLD interpretation/participant state/causal claims not all named.

D127. NIK-ST-H-001, with-BR, judge partial [GOOD]

claim: The hold-out validation method is sensitive to how the dataset is partitioned, and different partitionings can yield different results, which motivates the need for strict separation of training and test statistics.

source: m4_pubget_papers_bundle_00007.txt

judge reasoning: The reference context directly supports the first part of the claim: 'However, different partitioning methods lead to different results, requiring multiple grouping to ensure the stability and persuasiveness of the model's result.' This confirms that hold-out validation is sensitive to partitioning and different partitions yield different results. However, the context does not explicitly state that this sensitivity 'motivates the need for strict separation of training and test statistics.' Instead, it states that hold-out validation 'randomly divides the original data set into independent training and test set' to 'avoid the overfitting problems caused by traditional training and testing based on the same data.' The sensitivity to partitioning is presented as a problem that 'requiring multiple grouping' to ensure stability, not as the motivation for the initial strict separation itself. Thus, the support for the causal link in the second part of the claim is not direct.

grade note: partial correct: partitioning sensitivity supported; causal link to strict train/test separation not direct.

D128. NIK-ST-H-002, with-BR, judge partial [GOOD]

claim: Feature selection in fMRI classification commonly involves anatomical or functional masking of uninformative voxels before analysis, and the point at which this masking occurs relative to the CV split determines whether it constitutes leakage.

source: m4_pubget_papers_bundle_00032.txt

judge reasoning: The reference context states that 'In most cases, the whole-brain contrast images are masked to remove voxels that are uninformative (e.g. voxels in ventricles; all voxels that are not in a specific ROI). Sometimes, you may want to use a functional mask...', which supports the first part of the claim regarding anatomical or functional masking of uninformative voxels in feature selection. However, the context does not mention 'CV split' or 'leakage' in relation to the timing of this masking, which is a significant part of the claim.

grade note: partial correct: functional/anatomical masking supported; CV-split timing and leakage absent from context.

D129. NIK-ST-H-002, with-BR, judge partial [GOOD]

claim: The pre-selection of pipeline steps or features outside the optimization framework re-introduces user-dependent biases and invalidates the cross-validation.

source: m4_pubget_papers_bundle_00092.txt

judge reasoning: The reference context explicitly states that "the "pre-selection" of a subset of pipeline steps may re-introduce user-dependent biases into the pipeline optimization framework." This directly supports the first part of the claim regarding user-dependent biases. However, the context does not state that this pre-selection "invalidates the cross-validation." While cross-validation is mentioned, its invalidation due to pre-selection is not asserted. Therefore, the support for the full claim is partial.

grade note: partial correct: pre-selection re-introducing user bias directly supported; 'invalidates cross-validation' not asserted.

D130. NIK-ST-H-003, with-BR, judge partial [GOOD]

claim: Nested cross-validation, with an inner loop for hyperparameter optimization and an outer loop for performance estimation, is the standard unbiased evaluation scheme in neuroimaging ML.

source: m4_pubget_papers_bundle_00011.txt

judge reasoning: The reference context describes that 'the authors used a nested cross-validation scheme' for 'hyperparameter optimization (1) and performance estimation (2)', which aligns with the first part of the claim. However, the context does not state or imply that this method 'is the standard unbiased evaluation scheme in neuroimaging ML'. It only describes its application in a specific study, not its general status within the field.

grade note: partial correct: nested CV inner/outer loops described in one study; 'standard' field-wide status not supported.

D131. NIK-ST-H-006, with-BR, judge partial [GOOD]

claim: NHST pits a point null hypothesis against all alternatives and provides a restricted picture of effects, particularly when combined with conservative multiple testing corrections.

source: m4_pubget_papers_bundle_00058.txt

judge reasoning: The reference context directly states that the NHST framework "pits a point null hypothesis against all alternatives, and thus provides only a rather restricted picture of the effects in a data set." It also mentions that the "multiple testing problem that is addressed by applying conservative corrections to the critical value." While both elements of the claim (restricted picture and conservative multiple testing corrections) are present in the text, the context does not explicitly state that the restricted picture is *particularly* so *when combined with* these corrections. It presents them as separate, though related, issues within the fMRI context, rather than explicitly linking them as an exacerbating factor.

grade note: Partial is defensible: rationale notes source doesn't link 'restricted picture' with corrections as claim's 'particularly when combined' asserts.

D132. NIK-ST-H-009, with-BR, judge partial [GOOD]

claim: ROPE-based decisions in fMRI can be operationalized by combining a posterior probability threshold with HDI/ROPE overlap rules, and ROPE maps can summarize the maximum effect-size threshold at which a voxel is classified as activated, deactivated, or not activated.

source: m4_pubget_papers_bundle_00005.txt

judge reasoning: The first part of the claim, that 'ROPE-based decisions in fMRI can be operationalized by combining a posterior probability threshold with HDI/ROPE overlap rules,' is directly supported by the context: 'apply the posterior probability threshold $p = 95\%$ or consider the overlap between the 95% HDI and ROPE.' The second part of the claim states that 'ROPE maps can summarize the maximum effect-size threshold at which a voxel is classified as activated, deactivated, or not activated.' The context supports this for 'activated/deactivated' voxels ('the ROPE map contains the maximum ES thresholds that allow voxels to be classified as 'activated/deactivated)'). However, for 'not activated' voxels, the context states 'the map contains the minimum effect size thresholds that allow voxels to be classified as 'not activated,' which contradicts the claim's assertion of 'maximum effect-size threshold' for 'not activated' voxels. Due to this specific discrepancy, the support is partial.

grade note: partial well-reasoned: HDI/ROPE rules supported, but 'not activated' uses min not max threshold, a real contradiction.

D133. NIK-ST-M-005, with-BR, judge partial [GOOD]

claim: The effect of motion correction model aggressiveness on t-statistics depends on head motion severity (FD): more conservative models improve t-statistics when motion is high, but in low-motion subjects the benefit is reduced or reversed.

source: m4_pubget_papers_bundle_00052.txt

judge reasoning: The reference context supports the first part of the claim, stating that 'the Conservative model significantly increased t-statistics compared to Aggressive when the FD or adjusted R were high'. However, it contradicts the second part of the claim regarding low-motion subjects. The claim states that 'in low-motion subjects the benefit is reduced or reversed', while the context explicitly says, 'When motion was lower, both Moderate and Conservative models demonstrated improvements compared to Aggressive'. Thus, the context supports the general idea of dependency on motion severity and the benefit for high motion, but not the specific outcome for low motion.

grade note: partial defensible: high-motion benefit supported; low-motion claim contradicted, so mixed support justifies partial.

D134. NIK-ST-M-008, with-BR, judge partial [GOOD]

claim: The nonparametric testing literature for neuroimaging is dominated by permutation tests, which test exchangeability under the null hypothesis of statistical independence between data and explanatory variables.

source: m4_pubget_papers_bundle_00003.txt

judge reasoning: The reference context explicitly states that 'The literature on nonparametric tests for neuroimaging data is dominated by permutation tests'. It also states that 'with a permutation test, one tests the null hypothesis of exchangeability...'. Separately, the context mentions 'the null hypothesis of statistical independence between the biological data and the explanatory variable is rejected.' While the context provides the individual components (permutation tests dominate, permutation tests test

exchangeability under a null hypothesis, and statistical independence is a null hypothesis), it does not explicitly combine them to state that permutation tests test 'exchangeability under the null hypothesis of statistical independence between data and explanatory variables' as a single, direct assertion. Therefore, the support is partial.

grade note: partial defensible though somewhat harsh: all components present, only the combined single assertion is not verbatim.

D135. NIK-TR-H-005, with-BR, judge partial [GOOD]

claim: Checking pairwise variance time series and framewise displacement correlations is a standard QC approach to identify extreme values and artefacts in preprocessed fMRI data.

source: m4_pubget_papers_bundle_00009.txt

judge reasoning: The reference context describes using framewise displacement and DVARS (derivative of variance in BOLD signal) to identify outlier volumes, which is a standard QC approach for fMRI data. This supports the general idea of using these measures for QC. However, the claim specifically mentions 'pairwise variance time series and framewise displacement correlations' as the QC approach. While DVARS is related to variance, the context does not mention 'pairwise variance time series' nor the 'correlations' between these measures as a QC method, but rather thresholding them individually. Therefore, it supports a weaker or tangential version of the claim.

grade note: partial correct: FD/DVARS thresholding supported but claim's 'pairwise variance correlations' framing only tangentially matches.

D136. NIK-TR-H-005, with-BR, judge partial [GOOD]

claim: Pairwise variance and framewise displacement timeseries can be used to localize extreme values and check whether preprocessed signals remain correlated with motion or WM/CSF derivatives.

source: m4_pubget_papers_bundle_00009.txt

judge reasoning: The reference context states that 'Outlier volumes in which framewise displacement exceeded 1mm, the derivative of variance in BOLD signal across the brain (DVARS) exceeded the upper fence...' This directly supports the first part of the claim, that framewise displacement and a derivative of variance (related to pairwise variance) can be used to localize extreme values (outlier volumes). However, the context does not state that these specific timeseries (pairwise variance and framewise displacement) are used to 'check whether preprocessed signals remain correlated with motion or WM/CSF derivatives.' It mentions using motion regressors and time-series from white matter and ventricles as nuisance regressors to control for physiological noise, but this is a different process than using pairwise variance and framewise displacement to check for *remaining* correlations.

grade note: partial correct: FD/DVARS localize outliers supported; 'check remaining correlation with motion/WM/CSF' is a different process.

Judge verdict = no_unrelated

D137. NIK-BP-E-004, with-BR, judge no_unrelated [GOOD]

claim: Using the same data to select a subset, ROI, or analysis path and then re-analyzing it for inference is circular analysis or double dipping and can invalidate statistical inference.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The provided reference context discusses factors affecting the BOLD signal, the challenge of drawing unbiased conclusions, and 'reverse inference'. It does not mention 'circular analysis', 'double dipping', or the specific methodological issue of using the same data for selection and subsequent inference, nor does it discuss the invalidation of statistical inference due to such practices. Therefore, the context does not support the specific claim.

grade note: BOLD/reverse-inference context omits circular-analysis/double-dipping; no_unrelated defensible.

D138. NIK-BP-E-004, with-BR, judge no_unrelated [GOOD]

claim: Circular analysis (double dipping) arises when the same data are used for selection and subsequent analysis, violating independence between the test statistic and the selection criterion under the null hypothesis.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The provided reference context discusses fMRI research, BOLD signal, and 'reverse inference'. It does not mention 'circular analysis' or 'double dipping' or the definition provided in the claim regarding using the same data for selection and subsequent analysis, or the violation of independence.

grade note: Reverse-inference context lacks circular-analysis/double-dipping definition; consistent.

D139. NIK-BP-E-004, with-BR, judge no_unrelated [GOOD]

claim: Using the same data for selection and subsequent selective analysis (double dipping) results in invalid statistical inference by violating independence under the null hypothesis.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The provided reference context discusses factors affecting BOLD signals and the concept of 'reverse inference' in fMRI research. It does not contain any information about 'double dipping,' invalid statistical inference, or the violation of independence under the null hypothesis, which are the core components of the claim.

grade note: Context (BOLD factors, reverse inference) off-topic vs double-dipping claim. Correct.

D140. NIK-BP-H-001, with-BR, judge no_unrelated [GOOD]

claim: Differences between AFNI, FSL, and SPM results can be attributable to particular analysis stages, and the source of variability can differ by dataset.

source: m4_pubget_papers_bundle_00085.txt

judge reasoning: The claim specifically mentions 'AFNI, FSL, and SPM results', none of which are mentioned or discussed in the provided reference context. While the context discusses 'preprocessing pipeline' and 'processing choice', it does not link these to the specific software packages named in the claim. Furthermore, the context concludes that 'CPM findings observed here were not being

driven by a particular processing choice', which contradicts the idea that differences *can be attributable* to particular analysis stages for their specific findings. The claim's assertion that 'the source of variability can differ by dataset' is also not addressed in the context.

grade note: AFNI/FSL/SPM absent and context arguably contradicts claim; no_unrelated defensible though 'no' could also fit.

D141. NIK-BP-H-001, with-BR, judge no_unrelated [REVIEW]

claim: Variation between software packages (AFNI, FSL, SPM) in fMRI results is largely attributable to a handful of individual analysis stages, and these sources of variability are heterogeneous across datasets.

source: m4_pubget_papers_bundle_00085.txt

judge reasoning: The claim states that variation between specific software packages (AFNI, FSL, SPM) is attributable to analysis stages and is heterogeneous across datasets. The provided context discusses altering preprocessing pipelines (functional parcellation, global signal regression) and notes that 'the overall prediction patterns across scans remained consistent' and that 'CPM findings observed here were not being driven by a particular processing choice.' This contradicts the claim's assertion of heterogeneity across datasets regarding sources of variability. Furthermore, the context does not mention the specific software packages listed in the claim.

grade note: Rationale describes context contradicting the claim (heterogeneity vs robustness), which is 'no', not 'no_unrelated'. Mislabeled.

D142. NIK-BP-H-001, with-BR, judge no_unrelated [GOOD]

claim: Variation between software packages (AFNI, FSL, SPM) is largely attributable to specific analysis stages, particularly motion correction, smoothing, and normalization

source: m4_pubget_papers_bundle_00085.txt

judge reasoning: The claim states that variation between specific software packages (AFNI, FSL, SPM) is attributable to specific analysis stages (motion correction, smoothing, normalization). The provided reference context discusses the robustness of findings to preprocessing choices like functional parcellation and global signal regression, concluding that the findings were 'not being driven by a particular processing choice'. It does not mention AFNI, FSL, SPM, motion correction, smoothing, or normalization, nor does it discuss variation *between* software packages. Therefore, the context does not support the specific claim.

grade note: Context omits named packages and processing stages; framed as absence not contradiction. Unrelated defensible.

D143. NIK-BP-H-002, with-BR, judge no_unrelated [GOOD]

claim: Variation between software packages like AFNI, FSL, and SPM is largely attributable to specific analysis stages and is heterogeneous across datasets.

source: m4_pubget_papers_bundle_00085.txt

judge reasoning: The claim states that variation between specific software packages (AFNI, FSL, SPM) is attributable to analysis stages and heterogeneous across datasets. The provided reference context discusses the robustness of a study's findings to different preprocessing choices (functional parcellation, global signal regression) within their own pipeline. It does not mention AFNI, FSL, or SPM, nor does it discuss variation *between* software packages or heterogeneity *across datasets*. In fact, the context suggests consistency ('overall prediction patterns across scans remained consistent', 'not being driven by a particular processing choice'), which contrasts with the claim's assertion of variation and heterogeneity.

grade note: Context on within-pipeline robustness, never names AFNI/FSL/SPM; even contradicts claim. Unrelated defensible.

D144. NIK-BP-H-002, with-BR, judge no_unrelated [GOOD]

claim: The NARPS study demonstrated that 70 research teams produced different results using different pipelines on the same task-fMRI dataset

source: m4_pubget_papers_bundle_00010.txt

judge reasoning: The claim states that 'The NARPS study demonstrated that 70 research teams produced different results using different pipelines on the same task-fMRI dataset'. The provided reference context discusses the importance of understanding how different analysis strategies and pipeline choices influence statistical outcomes in fMRI analysis, and mentions using data from the Human Connectome Project. However, it does not mention the 'NARPS study' or '70 research teams' or their specific findings. Therefore, the context does not support the specific claim.

grade note: Pipeline/HCP context never names NARPS or 70 teams; unrelated defensible.

D145. NIK-BP-H-004, with-BR, judge no_unrelated [GOOD]

claim: Neuroimaging predictive models with small effect sizes are highly sensitive to variations in preprocessing, feature computation, and model choice, impacting their robustness and reliability.

source: m4_pubget_papers_bundle_00011.txt

judge reasoning: The claim states that neuroimaging predictive models with small effect sizes are highly sensitive to variations in preprocessing, feature computation, and model choice, impacting their robustness and reliability. The provided reference context discusses different combinations of models and parcellations, hyperparameter optimization, and the impact of machine learning source code updates on evaluation. While it touches upon variations in model choice and potentially feature computation (parcellation), it does not mention 'small effect sizes,' nor does it explicitly state that models are 'highly sensitive' to these variations, or that this sensitivity impacts 'robustness and reliability.' The context describes a methodology for evaluating different model aspects, but not the specific claim about sensitivity and its consequences.

grade note: Model/parcellation context lacks small-effect-size sensitivity/robustness framing; no_unrelated defensible, mildly harsh.

D146. NIK-BP-H-004, with-BR, judge no_unrelated [GOOD]

claim: Neuroimaging prediction models can be underpowered and sensitive to workflow variation when effect sizes are small.

source: m4_pubget_papers_bundle_00011.txt

judge reasoning: The provided reference context describes methodological details such as scikit-learn versions, parcellations, hyperparameter optimization, and cross-validation schemes. It does not contain any information about neuroimaging prediction

models being underpowered, sensitive to workflow variation specifically when effect sizes are small, or the concept of effect sizes at all. Therefore, it does not support the claim.

grade note: Methods-detail context lacks underpowered/effect-size sensitivity content; no_unrelated correct.

D147. NIK-BP-H-004, with-BR, judge no_unrelated [GOOD]

claim: Imaging features often contribute limited predictive value compared to clinical covariates, and model performance at certain time points is minimally affected by variations in preprocessing, feature computation, or model choice.

source: m4_pubget_papers_bundle_00011.txt

judge reasoning: The provided reference context discusses scikit-learn versions, parcellations, model combinations, hyperparameter optimization, and cross-validation. It does not contain any information about the predictive value of imaging features compared to clinical covariates, nor does it discuss how model performance is affected by variations in preprocessing, feature computation, or model choice. Therefore, the context does not support the claim.

grade note: sklearn/parcellation/CV context lacks imaging-vs-clinical predictive claim; unrelated defensible.

D148. NIK-BP-H-005, with-BR, judge no_unrelated [REVIEW]

claim: Multiverse analysis addresses the problem that the number of analysis steps combined with possible parameters for each step creates a vast multiplicity of methodologic variants.

source: m4_pubget_papers_bundle_00057.txt

judge reasoning: The reference context describes the problem that 'analytic flexibility combined with the number of analysis steps, as well as the possible parameters for running each analysis step, has led to a vast multiplicity of methodologic variants'. This aligns with the problem described in the claim. However, the claim states that 'Multiverse analysis addresses' this problem. The provided reference context does not mention 'Multiverse analysis' or any solution to the described problem, thus it does not support the specific claim.

grade note: Rationale quotes context stating the exact 'vast multiplicity of methodologic variants' problem the claim describes; substantial support, no_unrelated too harsh.

D149. NIK-BP-H-006, with-BR, judge no_unrelated [GOOD]

claim: Specification-curve analysis uses a rank-ordered visualization of effect estimates (e.g., median ICC coefficients with 95% CIs) paired with a panel of analytic decisions, enabling researchers to see which decisions drive variation across hundreds of model specifications.

source: m4_pubget_papers_bundle_00032.txt

judge reasoning: The provided reference context discusses fMRI data processing steps such as smoothing and feature selection (masking). It does not contain any information about 'specification-curve analysis', 'rank-ordered visualization', 'effect estimates', 'ICC coefficients', 'analytic decisions', or 'model specifications' as described in the claim. The context is entirely unrelated to the claim.

grade note: Smoothing/masking context lacks specification-curve/ICC content; no_unrelated correct.

D150. NIK-BP-H-006, with-BR, judge no_unrelated [GOOD]

claim: Hierarchical linear modeling can evaluate the effect of analytic decisions on outcome measures

source: m4_pubget_papers_bundle_00032.txt

judge reasoning: The provided reference context discusses smoothing, first-level and second-level analyses, and feature selection (masking) in neuroimaging. It does not mention 'Hierarchical linear modeling' or its use in evaluating the effect of analytic decisions on outcome measures. Therefore, the context does not support the claim.

grade note: Second-level analysis is loosely HLM-adjacent, but context never names HLM/analytic-decision evaluation; defensible.

D151. NIK-BP-M-001, with-BR, judge no_unrelated [GOOD]

claim: Preregistration is used to address reproducibility concerns, and reproducible analysis workflows are needed so results can be replicated within laboratories.

source: m4_pubget_papers_bundle_00095.txt

judge reasoning: The claim consists of two distinct parts. The first part, 'Preregistration is used to address reproducibility concerns,' is not supported by the provided reference context, as the term 'preregistration' is not mentioned anywhere in the excerpt. The second part, 'reproducible analysis workflows are needed so results can be replicated within laboratories,' is partially supported by the context's discussion of 'scripts of the whole analysis workflow, which should be learned for efficiency, reproducibility and provenance tracking.' However, because a significant and distinct component of the overall claim is entirely unsupported by the reference, the claim as a whole cannot be considered supported.

grade note: Rationale notes second half partially supported but first half (preregistration) absent; whole-claim no_unrelated is defensible, slightly harsh.

D152. NIK-BP-M-001, with-BR, judge no_unrelated [GOOD]

claim: For analyses that are exploratory or post-hoc, p-values should not be attached to the results.

source: m4_pubget_papers_bundle_00095.txt

judge reasoning: The provided reference context discusses fMRI analysis workflows, reproducibility, provenance tracking, and pipeline systems. It does not contain any information or recommendations regarding p-values, exploratory analyses, or post-hoc analyses, and therefore does not support the claim.

grade note: Workflow/reproducibility context lacks p-value/exploratory guidance; no_unrelated correct.

D153. NIK-BP-M-001, with-BR, judge no_unrelated [GOOD]

claim: Pre-registration is an effective solution to address reproducibility concerns and the file drawer issue in fMRI research

source: m4_pubget_papers_bundle_00095.txt

judge reasoning: The claim states that 'Pre-registration is an effective solution to address reproducibility concerns and the file drawer issue in fMRI research'. The provided reference context discusses the use of analysis workflow scripts and pipelines for efficiency, reproducibility, and provenance tracking in fMRI data processing. While it mentions 'reproducibility', it does not mention 'pre-registration' as a solution, nor does it address the 'file drawer issue'. Therefore, the context does not support the specific claim.

grade note: Only generic 'reproducibility' overlap; preregistration/file-drawer specifics absent, so no_unrelated defensible.

D154. NIK-BP-M-003, with-BR, judge no_unrelated [GOOD]

claim: Multivariate and model-based fMRI analyses can increase statistical power by reducing the multiple comparisons burden inherent in mass univariate approaches.

source: m4_pubget_papers_bundle_00065.txt

judge reasoning: The claim states that multivariate and model-based fMRI analyses can increase statistical power by reducing the multiple comparisons burden inherent in mass univariate approaches. The provided reference context discusses the general determinants of statistical power (sample size, effect size, Type I error rate) but does not mention fMRI analyses, multivariate or univariate approaches, or the multiple comparisons burden. Therefore, the context does not support the specific claim.

grade note: Generic power-determinants context, no multivariate/multiple-comparisons content; defensible.

D155. NIK-BP-M-003, with-BR, judge no_unrelated [GOOD]

claim: ROI analyses alone cannot quantify the specificity of identified effects; interpretation changes if 60% of the rest of the brain shows the same effect

source: m4_pubget_papers_bundle_00065.txt

judge reasoning: The provided reference context discusses statistical power, sample size, effect size, and Type I error rates in fMRI studies. It does not address the limitations of ROI analyses regarding the specificity of identified effects or how widespread activation across the brain might alter interpretation, which is the core of the claim.

grade note: Context is power/sample-size; claim is ROI specificity. Genuinely different topics, no_unrelated correct.

D156. NIK-BP-M-003, with-BR, judge no_unrelated [GOOD]

claim: ROI analyses alone cannot quantify specificity of identified effects; interpretation changes if many other regions show the same effect

source: m4_pubget_papers_bundle_00065.txt

judge reasoning: The claim discusses the limitations of ROI analyses in quantifying specificity and how interpretation changes if effects are widespread. The provided reference context, however, focuses on statistical power, sample size, effect size, and Type I error rates, and mentions amygdala activation in response to emotional faces. It does not contain any information related to ROI analyses, the specificity of effects, or the interpretation of findings across multiple brain regions.

grade note: Context is power/sample-size plus amygdala example; ROI-specificity claim unsupported. Correct.

D157. NIK-BP-M-003, with-BR, judge no_unrelated [GOOD]

claim: Multivariate and model-based fMRI approaches can increase statistical power partly by reducing the per-voxel/per-ROI multiple-comparison burden.

source: m4_pubget_papers_bundle_00065.txt

judge reasoning: The claim states that multivariate and model-based fMRI approaches can increase statistical power by reducing the multiple-comparison burden. The provided reference context discusses general principles of statistical power (sample size, effect size, Type I error rate) but does not mention multivariate or model-based fMRI approaches, nor does it discuss the concept of reducing multiple-comparison burden. Therefore, the context does not support the specific claim.

grade note: Context is general power principles; multivariate-approach/MC-burden claim absent. Verdict defensible.

D158. NIK-BP-M-004, with-BR, judge no_unrelated [GOOD]

claim: Pipeline variability from motion correction, smoothing kernel size, and model parameterization explains substantial variance in estimated coefficients

source: m4_pubget_papers_bundle_00012.txt

judge reasoning: The claim discusses pipeline variability in neuroimaging (motion correction, smoothing kernel size, model parameterization) and its effect on estimated coefficients. The provided reference context, however, discusses 'reverse inference' in cognitive neuroscience, specifically the challenges of inferring psychological processes from brain activity and the example of the anterior cingulate cortex. There is no mention of fMRI processing pipelines, variability, or estimated coefficients in the provided text. The context is entirely unrelated to the claim.

grade note: Reverse-inference context unrelated to pipeline-variability/coefficient variance; no_unrelated correct.

D159. NIK-BP-M-004, with-BR, judge no_unrelated [GOOD]

claim: Stringent motion correction decreases between-subject variance, with motion parameters explaining substantial variability in estimated coefficients across pipelines

source: m4_pubget_papers_bundle_00012.txt

judge reasoning: The claim discusses stringent motion correction, between-subject variance, motion parameters, and estimated coefficients in neuroimaging. The provided reference context, however, focuses entirely on 'reverse inference' in cognitive neuroscience and an example involving the anterior cingulate cortex and cognitive conflict. There is no overlap in subject matter; the context does not support the claim in any way.

grade note: Reverse-inference context vs motion-correction claim; genuinely unrelated.

D160. NIK-BP-M-004, with-BR, judge no_unrelated [GOOD]

claim: Across many pipelines, contrast choice, model parameterization, smoothing FWHM, and motion handling each explain non-trivial variance in estimated effects, motivating sensitivity reporting.

source: m4_pubget_papers_bundle_00012.txt

judge reasoning: The claim discusses factors influencing variance in estimated effects across neuroimaging pipelines (contrast choice, model parameterization, smoothing FWHM, motion handling). The provided reference context, however, focuses entirely on the concept of 'reverse inference' in cognitive neuroscience and the challenges of inferring psychological processes from brain activity. There is no mention of any of the specific topics or concepts presented in the claim within the provided context.

grade note: Reverse-inference context vs pipeline-variance claim; unrelated correct.

D161. NIK-BP-M-007, with-BR, judge no_unrelated [GOOD]

claim: Motion correction approaches utilizing frame censoring require quantification of motion artifacts through thresholding of metrics like FD and DVARS.

source: m4_pubget_papers_bundle_00088.txt

judge reasoning: The provided reference context discusses 'overlap maps' in fMRI studies, group analysis, and individual subject inferences, particularly in the context of brain damage. It contains no information about motion correction, frame censoring, motion artifacts, FD, or DVARS, which are the central topics of the claim. Therefore, the context is entirely unrelated to the claim.

grade note: Overlap-map context lacks frame censoring/FD/DVARS; no_unrelated justified.

D162. NIK-BP-M-008, with-BR, judge no_unrelated [GOOD]

claim: Reporting the output voxel size is critical; the resampling voxel size should be similar to the original acquisition voxel size.

source: m4_pubget_papers_bundle_00009.txt

judge reasoning: The provided reference context describes various fMRI preprocessing steps such as motion correction, skull-stripping, despiking, and smoothing with a 6mm kernel. It also mentions nuisance regressors and outlier detection. However, it does not contain any information or discussion about the importance of reporting output voxel size, resampling voxel size, or the need for resampling voxel size to be similar to the original acquisition voxel size. Therefore, the context does not support the claim.

grade note: Preprocessing-steps context omits voxel-size reporting/resampling; no_unrelated defensible.

D163. NIK-BP-M-008, with-BR, judge no_unrelated [GOOD]

claim: For SPM workflows, normalization parameters are obtained from segmentation and applied via Normalize:Write functionality with deformation field (y_XXX file) and voxel sizes set to match original acquisition resolution.

source: m4_pubget_papers_bundle_00009.txt

judge reasoning: The claim describes a specific normalization workflow using SPM, including details like 'segmentation', 'Normalize:Write functionality', 'deformation field (y_XXX file)', and 'voxel sizes set to match original acquisition resolution'. The provided reference context, however, explicitly states that preprocessing was performed using FSL and AFNI, and details steps such as motion correction, skull-stripping, despiking, and smoothing. There is no mention of SPM or any of the specific SPM-related normalization details from the claim in the context.

grade note: Context explicitly uses FSL/AFNI, not SPM Normalize:Write workflow; unrelated correct.

D164. NIK-BP-M-008, with-BR, judge no_unrelated [GOOD]

claim: Systematic normalization errors between groups can trigger false positive activations, while misalignment between individuals increases false negatives.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The claim discusses 'systematic normalization errors' and 'misalignment between individuals' leading to 'false positive activations' and 'false negatives'. The provided reference context discusses factors affecting the BOLD signal (age, medication, chemical compounds, respiration, oxygen level) and the concept of 'reverse inference' in fMRI. There is no mention of normalization errors, misalignment, false positives, or false negatives in the provided text. Thus, the context is unrelated to the specific claim.

grade note: Context (BOLD factors, reverse inference) off-topic vs normalization-error false-positive claim. Correct.

D165. NIK-BP-M-008, with-BR, judge no_unrelated [GOOD]

claim: Nonlinear versus linear normalization represents a fundamental methodological fork that must be explicitly reported, and template choice (MNI, Talairach) must be specified for coordinate reporting.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The provided reference context discusses factors affecting the BOLD signal and the concept of 'reverse inference' in fMRI research. It does not contain any information about nonlinear versus linear normalization, the need to report methodological choices, or the specification of templates like MNI or Talairach for coordinate reporting. Therefore, it does not support the claim.

grade note: Context (BOLD factors, reverse inference) off-topic vs normalization/template-reporting claim. Correct.

D166. NIK-IN-H-001, with-BR, judge no_unrelated [GOOD]

claim: Direct measurement of electrical activity (EEG) and/or blood flow is the most promising approach for separating neuronal contributions from neurovascular coupling or metabolic contributions to the BOLD signal

source: m4_pubget_papers_bundle_00004.txt

judge reasoning: The provided reference context discusses DVARS, BOLD signal changes related to head motion, and a study design involving oxytocin. It does not mention EEG, direct measurement of electrical activity, blood flow, or any approach for separating neuronal contributions from neurovascular coupling or metabolic contributions to the BOLD signal. The context is entirely unrelated to the claim.

grade note: Context (DVARS/motion/oxytocin) clearly off-topic vs EEG/neurovascular-coupling claim. Correct.

D167. NIK-IN-H-001, with-BR, judge no_unrelated [GOOD]

claim: Changes in neuromodulator concentrations can profoundly alter the relationships between the BOLD signal, blood flow, energy metabolism, and neuronal activity; if these relationships are not constant across brain regions and pathways, a control task or stimulus will never be sufficient to exclude the possibility that neurovascular coupling differences are responsible for BOLD signal differences.

source: m4_pubget_papers_bundle_00004.txt

judge reasoning: The provided reference context discusses head motion, DVARS, and a study design involving intranasal oxytocin. It mentions the BOLD signal in the context of measuring neural activation and its relationship to motion. However, it does not contain any information about how neuromodulator concentrations alter the relationships between the BOLD signal, blood flow, energy metabolism, and neuronal activity, nor does it discuss neurovascular coupling differences as described in the claim. The content of the context is unrelated to the specific claim.

grade note: Context (motion/DVARS/oxytocin) lacks neuromodulator/neurovascular-coupling content of claim. Defensible.

D168. NIK-IN-H-002, with-BR, judge no_unrelated [GOOD]

claim: The CO₂-driven BOLD response is a well-established method for mapping cerebrovascular reactivity (CVR) by relying on pathways like nitric oxide to increase arterial diameter.

source: m4_pubget_papers_bundle_00011.txt

judge reasoning: The claim discusses CO₂-driven BOLD response and cerebrovascular reactivity. The provided reference context, however, is entirely focused on machine learning, scikit-learn, parcellation, and cross-validation, with no mention of the topics in the claim.

grade note: Context is ML/scikit-learn/CV; claim is CO₂-BOLD/CVR. Clearly unrelated.

D169. NIK-IN-H-003, with-BR, judge no_unrelated [GOOD]

claim: Computational fMRI analyses require generating trial-by-trial parametric regressors (e.g., evidence contribution) while strictly controlling for EVs of no interest to avoid confounding variables.

source: m4_pubget_papers_bundle_00019.txt

judge reasoning: The reference context discusses the general challenges of analytic flexibility in neuroimaging, leading to false positives and non-reproducible results, and the need for judicious choices in analysis pipelines. However, it does not mention 'computational fMRI analyses,' 'trial-by-trial parametric regressors,' 'evidence contribution,' or 'controlling for EVs of no interest to avoid confounding variables.' The context is too general and does not support the specific requirements outlined in the claim.

grade note: Generic analytic-flexibility context vs specific parametric-regressor claim; unrelated defensible.

D170. NIK-IN-H-003, with-BR, judge no_unrelated [GOOD]

claim: Multivariate activity patterns can reveal more specific area-level processing roles and representational content than traditional univariate subtraction or parametric modulation.

source: m4_pubget_papers_bundle_00092.txt

judge reasoning: The provided reference context discusses the effects of preprocessing choices and pipeline optimization in fMRI data analysis. It does not contain any information about multivariate activity patterns, univariate subtraction, or parametric modulation, nor does it compare their efficacy in revealing area-level processing roles or representational content. Therefore, the context is unrelated to the claim.

grade note: Preprocessing-pipeline context vs multivariate-pattern claim; unrelated correct.

D171. NIK-IN-H-004, with-BR, judge no_unrelated [REVIEW]

claim: The global signal contains both non-neuronal confounds arising from motion, cardiac, and respiratory cycles, and fluctuations in widespread neuronal activity.

source: m4_pubget_papers_bundle_00031.txt

judge reasoning: The claim states what the global signal contains (non-neuronal confounds and neuronal activity). The provided reference context discusses the debate around Global Signal Regression (GSR) as a processing technique, its mathematical implications (anti-correlations), and its benefits or detriments for functional connectivity. It does not describe the components or content of the global signal itself. Therefore, the context does not support the claim.

grade note: GSR-debate context is directly about the global signal's content (neuronal vs confounds); calling it fully unrelated is too harsh, partial fits.

D172. NIK-IN-M-005, with-BR, judge no_unrelated [GOOD]

claim: Neurovascular coupling relationships between BOLD signal, blood flow, energy metabolism, and neuronal activity can be profoundly altered by disease state, neuromodulator concentrations, and behavioral context, making control tasks insufficient to rule out vascular explanations for BOLD differences.

source: m4_pubget_papers_bundle_00004.txt

judge reasoning: The reference context discusses head motion, DVARS, BOLD signal, and a study design involving oxytocin and mothers with addictions. While it mentions BOLD signal and factors like 'addictions' (disease state) and 'oxytocin' (neuromodulator), it does not discuss neurovascular coupling relationships, blood flow, energy metabolism, or how these relationships are altered by disease state, neuromodulator concentrations, or behavioral context. Crucially, it does not make the claim that control tasks are insufficient to rule out vascular explanations for BOLD differences. The context is real and parseable but does not support the specific claim.

grade note: Motion/oxytocin context touches BOLD but not neurovascular-coupling alteration claim; no_unrelated defensible.

D173. NIK-IN-M-005, with-BR, judge no_unrelated [GOOD]

claim: BOLD differences can arise from changes in neurovascular coupling or metabolic factors rather than neuronal activity alone.

source: m4_pubget_papers_bundle_00004.txt

judge reasoning: The provided reference context discusses head motion artifacts and their correlation with DVARS and BOLD signal changes, and how motion can alter observed neural activation patterns. It does not mention neurovascular coupling or metabolic factors as alternative sources for BOLD differences, nor does it discuss BOLD differences arising from factors other than neuronal activity alone. Therefore, the context is unrelated to the specific claim.

grade note: Motion-artifact context doesn't cover neurovascular-coupling/metabolic BOLD sources; no_unrelated defensible.

D174. NIK-IN-M-005, with-BR, judge no_unrelated [GOOD]

claim: Endogenous and exogenous factors including hormones, blood pressure, BMI, circadian rhythm, sleep, and age influence blood volume and flow parameters underlying the BOLD signal, and may cause a BOLD change while neuronal activity remains unchanged.

source: m4_pubget_papers_bundle_00033.txt

judge reasoning: The claim discusses various endogenous and exogenous factors influencing blood volume and flow parameters underlying the BOLD signal, and potentially causing BOLD changes independent of neuronal activity. The provided reference context, however, is entirely focused on technical aspects of fMRI data acquisition, specifically displacement fields, geometric distortions, head motion, and field map correction. It does not mention BOLD signal, blood volume, blood flow, or any of the factors listed in the claim. The content of the reference is unrelated to the claim.

grade note: Distortion/motion-correction context unrelated to BOLD physiological confounds; correct.

D175. NIK-IN-M-005, with-BR, judge no_unrelated [GOOD]

claim: Neurovascular coupling differences can produce BOLD differences that are not attributable to neural activity, so a control task alone cannot exclude this confound.

source: m4_pubget_papers_bundle_00004.txt

judge reasoning: The provided reference context discusses head motion artifacts in fMRI and a specific study design. It does not mention neurovascular coupling, how it might produce BOLD differences not attributable to neural activity, or the inability of a control task to exclude this confound. The context is unrelated to the specific claim.

grade note: Head-motion context vs neurovascular-coupling claim; genuinely unrelated.

D176. NIK-IN-M-005, with-BR, judge no_unrelated [GOOD]

claim: Direct measurement of electrical activity using EEG or baseline blood flow is recommended to separate neuronal activity contributions from neurovascular coupling confounds.

source: m4_pubget_papers_bundle_00004.txt

judge reasoning: The claim states that direct measurement of electrical activity (EEG) or baseline blood flow is recommended to separate neuronal activity from neurovascular coupling confounds. The provided reference context discusses head motion, DVARS, BOLD signal, and a study design involving oxytocin. It contains no information about EEG, baseline blood flow, neuronal activity contributions, or neurovascular coupling, and therefore does not support the claim.

grade note: Motion/oxytocin context vs EEG-recommendation claim; unrelated correct.

D177. NIK-IN-M-005, with-BR, judge no_unrelated [GOOD]

claim: BOLD signal depends on blood volume, blood flow, and oxygen extraction which can be influenced by endogenous factors causing BOLD changes without true neuronal activity differences

source: m4_pubget_papers_bundle_00033.txt

judge reasoning: The claim discusses the physiological basis of the BOLD signal, specifically its dependence on blood volume, blood flow, and oxygen extraction, and how endogenous factors can cause BOLD changes without true neuronal activity differences. The provided reference context, however, focuses entirely on fMRI data acquisition, displacement fields, geometric distortions, and head motion. It does not contain any information related to the BOLD signal's physiological underpinnings or the factors influencing it as described in the claim.

grade note: Context is acquisition/distortion/motion; claim is BOLD physiology (volume/flow/oxygen). Distinct subtopics, reasonable.

D178. NIK-IN-M-006, with-BR, judge no_unrelated [GOOD]

claim: Cognitive reserve theory distinguishes neural reserve (efficient use of same networks) from neural compensation (recruitment of alternative networks or strategies), and hyperactivation may reflect either adaptive compensation or maladaptive dedifferentiation

source: m4_pubget_papers_bundle_00006.txt

judge reasoning: The provided reference context discusses fMRI quality control methods, data acquisition, and preprocessing. It does not contain any information about cognitive reserve theory, neural reserve, neural compensation, hyperactivation, or dedifferentiation, which are the central topics of the claim.

grade note: QC/preprocessing context has no cognitive-reserve theory; no_unrelated justified.

D179. NIK-IN-M-006, with-BR, judge no_unrelated [GOOD]

claim: Increased recruitment in older adults is theoretically ambiguous between neural compensation and reduced neural efficiency/dedifferentiation, so the inefficiency account must be explicitly ruled out.

source: m4_pubget_papers_bundle_00006.txt

judge reasoning: The claim discusses the theoretical ambiguity of increased neural recruitment in older adults, specifically regarding neural compensation versus reduced neural efficiency/dedifferentiation, and the need to rule out the inefficiency account. The provided reference context, however, is entirely focused on fMRI data quality control methods, reporting standards, and the value of qualitative assessment in fMRI data. There is no mention of older adults, neural recruitment, compensation, efficiency, or dedifferentiation, making the context completely unrelated to the claim.

grade note: QC-methods context vs older-adult recruitment/compensation claim; clearly unrelated.

D180. NIK-IN-M-008, with-BR, judge no_unrelated [GOOD]

claim: ROI analyses increase sensitivity by averaging signal across voxels but require proper multiple comparisons correction within the ROI space.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The provided reference context discusses factors affecting the BOLD signal and the concept of 'reverse inference' in fMRI research. It does not contain any information about ROI analyses, averaging signal across voxels, increased sensitivity, or the requirement for multiple comparisons correction within ROI space, which are the core components of the claim.

grade note: Context (BOLD factors, reverse inference) off-topic vs ROI-averaging/MC-correction claim. Correct.

D181. NIK-IN-M-008, with-BR, judge no_unrelated [GOOD]

claim: ROI analyses can increase sensitivity by averaging over voxels but can also create a false sense of regional specificity.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The claim discusses the dual effects of ROI analyses: increasing sensitivity by averaging over voxels and creating a false sense of regional specificity. The provided reference context discusses the variability of the BOLD signal, challenges in fMRI research, and the concept of 'reverse inference' related to interpreting regional activation. However, it does not mention 'ROI analyses' or the specific mechanisms described in the claim (averaging over voxels for sensitivity or creating a false sense of regional specificity). While the context touches on issues with interpreting regional activation, it does not directly support the specific claim about ROI analyses.

grade note: no_unrelated correct: context on BOLD variability/reverse inference, never mentions ROI analyses or the claimed averaging/specificity mechanism.

D182. NIK-ME-H-001, with-BR, judge no_unrelated [GOOD]

claim: GE sequences emphasize BOLD contributions from larger veins, while SE acquisitions refocus these macrovascular contributions but retain microvascular sensitivity through diffusion-mediated dephasing

source: m4_pubget_papers_bundle_00014.txt

judge reasoning: The claim discusses the characteristics of GE and SE sequences in fMRI regarding BOLD contributions from different vascular compartments. The provided reference context, however, focuses on mock scanning, participant familiarity, stress/anxiety markers, developmental processes, and cognitive strategies in study design. There is no mention of fMRI sequences, BOLD, or vascular contributions in the context, making it entirely unrelated to the claim.

grade note: Mock-scanning context vs GE/SE sequence claim; clearly unrelated.

D183. NIK-ME-H-001, with-BR, judge no_unrelated [GOOD]

claim: The relative GE-BOLD and SE-BOLD contributions depend on vessel size, field strength, susceptibility, and diffusion-related spatial scales.

source: m4_pubget_papers_bundle_00014.txt

judge reasoning: The provided reference context discusses participant preparation, developmental processes, and cognitive strategies in MRI studies. This content is entirely unrelated to the claim about GE-BOLD and SE-BOLD contributions, vessel size, field strength, susceptibility, and diffusion-related spatial scales. Therefore, the context does not support the claim.

grade note: Context (participant prep/development) off-topic vs GE/SE-BOLD physics claim. Correct.

D184. NIK-ME-H-002, with-BR, judge no_unrelated [GOOD]

claim: The relative R2/R2* contributions depend on vessel size, field strength, and intravascular susceptibility, which is why field strength matters for SE microvascular specificity.

source: m4_pubget_papers_bundle_00014.txt

judge reasoning: The provided reference context discusses mock scanning, participant familiarity with the MRI environment, and cognitive development, which is entirely unrelated to the claim about R2/R2* contributions, vessel size, field strength, intravascular susceptibility, and SE microvascular specificity.

grade note: Mock-scanning context vs R2/R2* field-strength claim; clearly unrelated.

D185. NIK-ME-H-004, with-BR, judge no_unrelated [GOOD]

claim: PMC has been evaluated in task-free and task fMRI settings under different motion and resolution conditions.

source: m4_pubget_papers_bundle_00094.txt

judge reasoning: The claim states that 'PMC has been evaluated in task-free and task fMRI settings under different motion and resolution conditions.' The provided reference context describes the testing of hypotheses related to risk and impulsivity measures and their associations with fMRI patterns, using similarity measures. It does not mention 'PMC', nor does it discuss evaluation in task-free/task fMRI settings, or under different motion/resolution conditions. The context is entirely unrelated to the specific claim.

grade note: Context on risk/impulsivity fMRI never mentions PMC or task-free/motion/resolution eval; no_unrelated defensible. Offline 'yes' is wrong.

D186. NIK-ME-H-004, with-BR, judge no_unrelated [GOOD]

claim: PMC dynamically updates RF excitation and gradient waveforms to realign the field-of-view in real time with the subject's head movement.

source: m4_pubget_papers_bundle_00094.txt

judge reasoning: The claim discusses PMC, RF excitation, gradient waveforms, field-of-view realignment, and head movement. The provided reference context discusses hypotheses testing, similarity measures, correlation patterns, risk, impulsivity, and fMRI patterns. There is no overlap in content; the reference context does not support any part of the claim.

grade note: Risk/impulsivity context never covers PMC RF/gradient realignment; no_unrelated defensible.

D187. NIK-ME-H-004, with-BR, judge no_unrelated [GOOD]

claim: Increasing the number of motion nuisance parameters in post-hoc models can over-fit the data and reduce sensitivity to underlying neural activity.

source: m4_pubget_papers_bundle_00013.txt

judge reasoning: The claim states that 'Increasing the number of motion nuisance parameters in post-hoc models can over-fit the data and reduce sensitivity to underlying neural activity.' The provided reference context mentions 'subject motion' as a source of noise and discusses 'modelling approach, such as choice of haemodynamic response function (HRF), choice of regressors'. While it touches upon motion and modeling, it does not mention 'increasing the number of motion nuisance parameters', 'over-fitting the data',

or 'reduce sensitivity to underlying neural activity'. The context is too general and does not support the specific mechanism or outcome described in the claim.

grade note: Context mentions motion/regressors generically but not overfitting/sensitivity mechanism; no_unrelated defensible.

D188. NIK-ME-H-004, with-BR, judge no_unrelated [GOOD]

claim: PMC dynamically updates the sequence's RF excitation and gradient waveforms such that the field-of-view is realigned to match the subject's head movement.

source: m4_pubget_papers_bundle_00094.txt

judge reasoning: The claim discusses PMC dynamically updating RF excitation and gradient waveforms for field-of-view realignment to match head movement. The provided reference context, however, describes statistical methods for testing hypotheses related to similarity measures of fMRI patterns, risk, and impulsivity. There is no mention of PMC, RF excitation, gradient waveforms, field-of-view, or head movement in the context. Therefore, the context does not support the claim.

grade note: Similarity-measure context never covers PMC/FOV realignment; no_unrelated defensible.

D189. NIK-ME-H-004, with-BR, judge no_unrelated [REVIEW]

claim: Retrospective motion correction models with many parameters can over-fit low-motion subjects, reducing sensitivity to underlying neural activity.

source: m4_pubget_papers_bundle_00013.txt

judge reasoning: The reference context discusses general sources of noise in fMRI, including subject motion, and mentions modeling approaches. However, it does not contain any information about 'retrospective motion correction models with many parameters,' 'over-fitting low-motion subjects,' or how this might 'reduce sensitivity to underlying neural activity.' The provided text is too general and does not support the specific claim.

grade note: Context covers subject motion and noise modeling, same subtopic as claim; 'too general but related' fits partial, not no_unrelated.

D190. NIK-ME-H-004, with-BR, judge no_unrelated [GOOD]

claim: PMC dynamically updates RF excitation and gradient waveforms to realign the field-of-view to match subject head movement during acquisition

source: m4_pubget_papers_bundle_00094.txt

judge reasoning: The provided reference context discusses the testing of hypotheses related to similarity measures of fMRI patterns, risk, and impulsivity. It contains no information about PMC, RF excitation, gradient waveforms, field-of-view realignment, or subject head movement, which are the subjects of the claim. Therefore, the context is entirely unrelated to the claim.

grade note: Context (similarity measures/impulsivity) clearly off-topic vs PMC/RF/gradient claim. Correct.

D191. NIK-NK-E-003, with-BR, judge no_unrelated [REVIEW]

claim: Clusterwise inference in widely used fMRI software can produce false-positive rates up to 70% due to incorrect spatial autocorrelation distribution assumptions, undermining activation-based reverse-inference claims.

source: m4_pubget_papers_bundle_00025.txt

judge reasoning: The claim discusses specific issues with clusterwise inference in fMRI software, including false-positive rates up to 70% due to incorrect spatial autocorrelation assumptions, and its impact on reverse-inference claims. The provided reference context mentions rs-fMRI data and the goal of empirically measuring the capability of controlling FWER (Family-Wise Error Rate) when no true signal is present. While FWER is related to false positives in fMRI, the context does not mention 'clusterwise inference,' 'widely used fMRI software,' a '70% false-positive rate,' 'incorrect spatial autocorrelation distribution assumptions,' or 'undermining activation-based reverse-inference claims.' Therefore, the context is real and parseable but does not support the specific details of the claim.

grade note: Context empirically measures FWER control in rs-fMRI with no true signal, that is exactly false-positive/clusterwise territory; no_unrelated too harsh, should be partial.

D192. NIK-NK-H-001, with-BR, judge no_unrelated [GOOD]

claim: Stress and related physiological or behavioral state changes can reconfigure large-scale networks and confound resting-state fMRI recordings.

source: m4_pubget_papers_bundle_00069.txt

judge reasoning: The provided reference context discusses model-based fMRI data analysis, specifically focusing on models of neuronal populations, Bayesian model comparison, prediction error time-courses, and GLM approaches. It does not contain any information about stress, physiological or behavioral state changes, large-scale networks, or how these might reconfigure networks or confound resting-state fMRI recordings. The content is entirely unrelated to the claim.

grade note: Model-based fMRI context has nothing on stress/state reconfiguring networks; no_unrelated correct. Offline 'yes' is the wrong one.

D193. NIK-NK-H-003, with-BR, judge no_unrelated [GOOD]

claim: A decoder can identify information correlated with a construct while still not revealing the actual locally encoded features or tuning functions.

source: m4_pubget_papers_bundle_00092.txt

judge reasoning: The claim discusses decoders, identifying information correlated with a construct, and not revealing encoded features or tuning functions. The provided reference context, however, focuses entirely on fMRI preprocessing choices, pipeline effects, and cross-validation models. There is no mention of decoders, encoded features, or tuning functions in the context, making it unrelated to the claim.

grade note: Context is preprocessing/CV; claim is decoder tuning functions. Genuinely unrelated.

D194. NIK-PP-H-003, with-BR, judge no_unrelated [GOOD]

claim: Surface-based and volumetric analyses differ in analysis space size and statistical correction thresholds, with VBA typically having fewer comparisons than SBAV

source: m4_pubget_papers_bundle_00050.txt

judge reasoning: The claim discusses differences between surface-based and volumetric analyses, including analysis space size and statistical correction thresholds. The provided reference context, however, focuses on issues of inflating classification results by treating longitudinal scans as independent data points and does not mention any of the concepts or terms related to the claim (e.g., surface-based analysis, volumetric analysis, analysis space size, statistical correction thresholds).

grade note: Context on scan-independence inflation is unrelated to SBAV/VBA comparison; no_unrelated correct.

D195. NIK-PP-H-006, with-BR, judge no_unrelated [REVIEW]

claim: Susceptibility-induced geometric distortions in fMRI EPI data manifest as voxel compression at tissue-air boundaries and are attenuated after TOPUP correction.

source: m4_pubget_papers_bundle_00033.txt

judge reasoning: The reference context discusses 'geometric distortions' and 'displacement fields' in fMRI, and mentions limitations of 'most distortion correction approaches'. However, it does not mention 'susceptibility-induced', 'voxel compression at tissue-air boundaries', or 'TOPUP correction'. Therefore, the context does not support the specific details of the claim.

grade note: Context discusses geometric distortions and distortion correction, directly on-topic; missing 'TOPUP/susceptibility' wording warrants partial, not no_unrelated.

D196. NIK-PP-M-001, with-BR, judge no_unrelated [GOOD]

claim: Consensus has been reached that GSR appropriateness and usefulness depend on the dataset and scientific question, and the global signal frequently arises from motion-related and physiological (mainly respiratory) effects with spatially varying correlations across brain tissues.

source: m4_pubget_papers_bundle_00088.txt

judge reasoning: The claim discusses Global Signal Regression (GSR), its appropriateness, usefulness, and origins from motion and physiological effects. The provided reference context, however, focuses on overlap maps, standard group analysis, and the differences between group and individual level inferences in fMRI studies. There is no mention of GSR, global signal, motion, or physiological effects in the provided context, making it entirely unrelated to the claim.

grade note: Overlap-map context has no GSR/global-signal content; no_unrelated correct. Both judges agree.

D197. NIK-PP-M-001, with-BR, judge no_unrelated [GOOD]

claim: The global signal frequently arises from motion-related and physiological (mainly respiratory) effects, but also correlates positively with band-limited power from local field potentials across the cortex, confirming a neural component.

source: m4_pubget_papers_bundle_00088.txt

judge reasoning: The provided reference context discusses 'overlap maps' and the limitations of 'standard group analysis' in fMRI studies, emphasizing individual variability. It does not contain any information about the global signal, its origins (motion-related, physiological, respiratory), or its correlation with band-limited power from local field potentials or neural components, which are the subjects of the claim.

grade note: no_unrelated correct: context on overlap maps/individual variability, unrelated to global signal origins or LFP correlation.

D198. NIK-PP-M-002, with-BR, judge no_unrelated [GOOD]

claim: Confound parameter configurations in fMRIPrep derivatives (e.g., 24P, 36P, ACompCor, AROMA, custom confounds) define the nuisance regressor set and must be compatible with the temporal filtering applied during preprocessing.

source: m4_pubget_papers_bundle_00003.txt

judge reasoning: The provided reference context discusses nonparametric tests, permutation tests, and blocked designs in neuroimaging. It does not contain any information about fMRIPrep derivatives, specific confound parameters (e.g., 24P, 36P, ACompCor, AROMA), nuisance regressors, or the compatibility of these with temporal filtering, which are the core elements of the claim. Therefore, the context does not support the claim.

grade note: Nonparametric/permutation context vs fMRIPrep confound-config claim; unrelated defensible.

D199. NIK-ST-E-003, with-BR, judge no_unrelated [GOOD]

claim: Sharing unthresholded statistical maps supports interpretation by revealing underlying patterns such as functional networks, vascular artifacts, or physiological pulsations

source: m4_pubget_papers_bundle_00006.txt

judge reasoning: The provided reference context discusses fMRI QC methods, reporting standards, and the value of qualitative assessment in fMRI data. It does not mention 'unthresholded statistical maps' or how sharing them supports interpretation by revealing specific patterns such as functional networks, vascular artifacts, or physiological pulsations. Therefore, the context does not support the specific claim.

grade note: QC/reporting context lacks unthresholded-map interpretation claims; no_unrelated defensible.

D200. NIK-ST-E-003, with-BR, judge no_unrelated [GOOD]

claim: Sharing publicly available unthresholded maps supports interpretation by highlighting functional networks, biological effects, and potential artifacts.

source: m4_pubget_papers_bundle_00006.txt

judge reasoning: The claim is about the benefits of 'sharing publicly available unthresholded maps' for interpretation. The provided reference context discusses the need for consensus on QC approaches and revisiting reporting standards to improve cross-study interpretation and replicability, and mentions qualitative assessment and visual inspection. However, it does not mention 'unthresholded maps' or the specific benefits of highlighting 'functional networks, biological effects, and potential artifacts' as stated in

the claim. While the context generally relates to improving interpretation in fMRI, it does not support the specific mechanism or benefits described in the claim.

grade note: Context on QC/reporting standards, never mentions unthresholded maps or the specific benefits; no_unrelated is consistent.

D201. NIK-ST-H-003, with-BR, judge no_unrelated [GOOD]

claim: Classifying scans of the same subject as independent data points can inflate accuracy from 61% to 86%, and exploiting subject identifiability with 50 training subjects can achieve variance explained equivalent to 10,000 subjects without double-dipping.

source: fpsyt.2019.00983

judge reasoning: The claim discusses methodological issues in neuroimaging, specifically regarding data independence, accuracy inflation, and subject identifiability, citing specific percentages and subject counts. The provided reference context (title: "Abnormal Functional and Structural Connectivity of Amygdala-Prefrontal Circuit in First-Episode Adolescent Depression: A Combined fMRI and DTI Study") describes a specific empirical study on depression and brain connectivity. The title of the reference does not contain any information related to the methodological points or the specific numbers mentioned in the claim.

grade note: Only reference title available (depression connectivity study); unrelated to identifiability/accuracy-inflation claim. Reasonable given title.

D202. NIK-ST-H-004, with-BR, judge no_unrelated [GOOD]

claim: Skewed clinical outcome distributions and floor effects limit the variance needed for reliable brain-behavior prediction and can explain null results.

source: m4_pubget_papers_bundle_00025.txt

judge reasoning: The provided reference context discusses fMRI regions, psychosis, schizophrenia, and controlling FWER in rs-fMRI analysis. It does not contain any information about skewed clinical outcome distributions, floor effects, their impact on variance for brain-behavior prediction, or their ability to explain null results, as stated in the claim.

grade note: Context (psychosis/schizophrenia/FWER) off-topic vs floor-effects/variance claim. Correct.

D203. NIK-ST-H-004, with-BR, judge no_unrelated [GOOD]

claim: Skewed clinical or behavioral outcome distributions and limited variability can weaken or complicate interpretation of predictive performance.

source: m4_pubget_papers_bundle_00025.txt

judge reasoning: The provided reference context discusses fMRI data, psychosis, schizophrenia, and statistical control (FWER). It does not contain any information about skewed clinical or behavioral outcome distributions, limited variability, or their impact on the interpretation of predictive performance. The content is unrelated to the specific claim.

grade note: Context (psychosis/schizophrenia/FWER) off-topic vs skewed-outcome predictive claim. Correct.

D204. NIK-ST-H-006, with-BR, judge no_unrelated [GOOD]

claim: Power calculations in fMRI require specifying mean activation, BOLD variance, Type I error, and sample size, supported by tools such as PowerMap, fMRIPower, and NeuroPower.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The provided reference context discusses factors affecting the BOLD signal and the concept of 'reverse inference' in fMRI. It does not contain any information about power calculations, the parameters required for them (mean activation, BOLD variance, Type I error, sample size), or specific tools for power analysis in fMRI such as PowerMap, fMRIPower, or NeuroPower. Therefore, the context is unrelated to the claim.

grade note: Reverse-inference context has no power-calculation tools; verdict consistent.

D205. NIK-ST-H-006, with-BR, judge no_unrelated [GOOD]

claim: Statistical power is critical for equivalence testing; underpowered studies cannot distinguish true equivalence from inability to detect meaningful effects, and BOLD response variance uncertainty complicates power estimation

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The provided reference context discusses factors influencing the BOLD signal and challenges in fMRI research, specifically regarding drawing conclusions and 'reverse inference'. However, it does not contain any information about 'statistical power', 'equivalence testing', 'underpowered studies', or 'power estimation', which are the core components of the claim.

grade note: Context is BOLD factors/reverse inference; claim is power/equivalence testing. Main claim unsupported, reasonable.

D206. NIK-ST-H-007, with-BR, judge no_unrelated [REVIEW]

claim: In underpowered fMRI studies, power remains extremely low even with substantial sample sizes, and significant findings identify only a small fraction of genuine activations with highly misleading effect size estimates

source: m4_pubget_papers_bundle_00065.txt

judge reasoning: The provided context discusses the general determinants of statistical power (sample size, effect size, Type I error rate) and the uncertainty surrounding effect size estimates. However, it does not state that power remains extremely low even with substantial sample sizes in underpowered fMRI studies, nor does it claim that significant findings identify only a small fraction of genuine activations or that effect size estimates are highly misleading in such studies. The context is related to the general topic but does not support the specific claims made.

grade note: Context is literally about statistical power, effect size, and estimate uncertainty, same topic. no_unrelated too harsh; should be partial.

D207. NIK-ST-H-007, with-BR, judge no_unrelated [REVIEW]

claim: Underpowered analyses may reveal significant activations that appear to tell a compelling story when true effects are actually weak and widely distributed.

source: m4_pubget_papers_bundle_00065.txt

judge reasoning: The claim states that 'Underpowered analyses may reveal significant activations that appear to tell a compelling story when true effects are actually weak and widely distributed.' The provided reference context discusses the definition of statistical power and its determinants (sample size, effect size, Type I error rate). It does not, however, discuss the specific consequence of underpowered analyses leading to misleading significant activations or 'compelling stories' when true effects are weak and distributed. While the context is related to statistical power, it does not support the specific claim made.

grade note: Context is on statistical power/significance, same topic as the underpowered-analysis claim; no_unrelated arguably too harsh, could be partial.

D208. NIK-ST-H-008, with-BR, judge no_unrelated [GOOD]

claim: The posterior odds are obtained by multiplying the Bayes Factor by the prior odds, so BF is a multiplicative factor updating prior beliefs rather than a direct posterior probability.

source: m4_pubget_papers_bundle_00077.txt

judge reasoning: The provided reference context discusses fMRI study preregistration and methodology standards in clinical trials. It does not contain any information about Bayes Factors, posterior odds, prior odds, or Bayesian inference, which are the subjects of the claim.

grade note: Preregistration/methodology context has no Bayes Factor content; no_unrelated correct. Both judges agree.

D209. NIK-ST-H-008, with-BR, judge no_unrelated [GOOD]

claim: $BF_{10} < 1$ indicates evidence favors the null hypothesis; $BF_{10} = 0.1$ means data support H_0 ten times more strongly than H_1

source: m4_pubget_papers_bundle_00007.txt

judge reasoning: The provided reference context discusses hold-out validation, data splitting, and model classification performance. It contains no information about Bayes Factors (BF_{10}), null hypothesis (H_0), or alternative hypothesis (H_1), which are the subjects of the claim. Therefore, the context does not support the claim.

grade note: Context (hold-out validation/classification) off-topic vs Bayes-factor claim. Correct.

D210. NIK-ST-H-009, with-BR, judge no_unrelated [GOOD]

claim: A data-driven ROPE can be defined by first selecting null voxels whose 50% HDI includes zero, then setting the ROPE as the 89% credible interval of that null-voxel population, with the 50% interval being an arbitrary choice.

source: m4_pubget_papers_bundle_00087.txt

judge reasoning: The claim describes a method for defining a data-driven ROPE using statistical concepts like 50% HDI and 89% credible intervals. The provided reference context, however, discusses fMRI data processing techniques, specifically CompCor variants, BOLD time-series, and nuisance regression. There is no mention of ROPE, HDI, credible intervals, or any related statistical concepts in the reference context. The '50 percent of variance' in the context is unrelated to '50% HDI' in the claim.

grade note: Context is CompCor/nuisance regression; ROPE/HDI/CI statistics absent. Rationale correctly distinguishes '50% variance' vs '50% HDI'.

D211. NIK-ST-M-001, with-BR, judge no_unrelated [GOOD]

claim: Averaging estimates over voxels within an ROI is an alternative to SVC that increases sensitivity but may create a false sense of specificity by masking activity patterns outside the ROI.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The provided document context discusses factors affecting the BOLD signal and the concept of 'reverse inference' in fMRI. It does not contain any information about averaging estimates over voxels within an ROI, SVC, sensitivity, specificity, or masking activity patterns, which are all central to the claim.

grade note: Reverse-inference context lacks ROI-averaging/SVC/specificity; no_unrelated correct.

D212. NIK-ST-M-001, with-BR, judge no_unrelated [GOOD]

claim: Restricting analysis to predefined ROIs (e.g., by averaging estimates or using Small-Volume Correction) limits the search space and increases statistical sensitivity compared to unconstrained whole-brain tests.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The provided reference context discusses factors affecting the BOLD signal and the concept of 'reverse inference' in fMRI research. It does not contain any information about restricting analysis to predefined ROIs, limiting search space, increasing statistical sensitivity, or comparing ROI analysis to whole-brain tests. Therefore, the context does not support the claim.

grade note: Reverse-inference context lacks ROI/SVC sensitivity comparison; no_unrelated correct.

D213. NIK-ST-M-001, with-BR, judge no_unrelated [GOOD]

claim: Small-volume correction is a recognized ROI-based approach that restricts voxelwise inference to the ROI search volume to reduce the multiple-comparisons burden.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The provided reference context discusses factors affecting BOLD signals and the concept of 'reverse inference' in fMRI research. It does not mention 'small-volume correction', 'ROI-based approach', 'voxelwise inference', or 'multiple-comparisons burden', which are all key components of the claim. Therefore, the context is unrelated to the claim.

grade note: no_unrelated correct: context lacks small-volume correction, ROI, voxelwise inference, multiple-comparisons content.

D214. NIK-ST-M-002, with-BR, judge no_unrelated [GOOD]

claim: Defining an ROI on the very basis of a statistical mapping of functional activation from the same data is an example of double dipping.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The provided reference context discusses challenges in fMRI research, reverse inference, and factors affecting the BOLD signal. It does not contain any information about 'double dipping' or the practice of defining ROIs based on statistical mapping from the same data, which is the subject of the claim. Therefore, the context does not support the claim.

grade note: Context on reverse inference/BOLD lacks any double-dipping discussion; no_unrelated justified.

D215. NIK-ST-M-002, with-BR, judge no_unrelated [GOOD]

claim: The relevant independence criterion is that the final test statistic should be independent of the selection criterion under the null hypothesis.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The provided reference context discusses factors affecting the BOLD signal and the concept of 'reverse inference' in fMRI research. It does not contain any information about 'independence criterion', 'final test statistic', 'selection criterion', or 'null hypothesis' as stated in the claim. Therefore, the context does not support the claim.

grade note: BOLD/reverse-inference context lacks independence-criterion/null-hypothesis content; no_unrelated correct.

D216. NIK-ST-M-002, with-BR, judge no_unrelated [GOOD]

claim: Using the same data to select an ROI and then re-analyze that selected ROI is circular analysis and yields invalid statistical inference.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The provided reference context discusses factors affecting the BOLD signal, the challenge of drawing unbiased conclusions in fMRI, and 'reverse inference'. However, it does not mention 'circular analysis', 'ROI selection', or the specific problem of using the same data for ROI selection and subsequent re-analysis leading to invalid statistical inference. Therefore, the context does not support the specific claim.

grade note: No circular-analysis/ROI content in reverse-inference context; verdict consistent.

D217. NIK-ST-M-002, with-BR, judge no_unrelated [GOOD]

claim: Circular analysis is unfortunately seen with some frequency in functional neuroimaging studies.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The provided reference context discusses challenges in fMRI research, specifically detailing the issue of 'reverse inference'. However, it does not mention 'circular analysis' or any related concept, nor does it discuss the frequency of such an issue in functional neuroimaging studies. Therefore, the context does not support the specific claim.

grade note: Same doc, no mention of circular analysis or its frequency; no_unrelated defensible.

D218. NIK-ST-M-002, with-BR, judge no_unrelated [GOOD]

claim: Double dipping in fMRI occurs when the same data are used to define a ROI (e.g., by thresholding a statistical map) and then re-analyze the selected voxels for the same contrast, producing invalid statistical inference because the test statistic and selection criterion are not independent under the null hypothesis.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The provided reference context discusses factors affecting the BOLD signal and defines 'reverse inference' in fMRI. It does not contain any information about 'double dipping', its definition, or the statistical inference problems it causes, as described in the claim.

grade note: no_unrelated correct: context defines reverse inference, contains nothing on double dipping or its statistical invalidity.

D219. NIK-ST-M-003, with-BR, judge no_unrelated [GOOD]

claim: Using the same data for selection and selective analysis (double dipping) violates the independence of test statistics from selection criteria under the null and yields invalid inference.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The claim discusses 'double dipping' and its impact on statistical independence and inference validity. The provided reference context discusses factors affecting BOLD signals, the challenge of drawing unbiased conclusions in fMRI, and 'reverse inference'. It does not mention 'double dipping' or the specific statistical issues described in the claim.

grade note: BOLD/reverse-inference context omits double-dipping independence issue; no_unrelated defensible.

D220. NIK-ST-M-003, with-BR, judge no_unrelated [GOOD]

claim: Defining an ROI from the very statistical map used to test a functional activation is a canonical example of circular analysis in fMRI.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The provided reference context discusses factors affecting the BOLD signal and introduces the concept of 'reverse inference' in fMRI. It does not mention 'circular analysis' or the specific scenario of defining an ROI from the same statistical map used for testing functional activation. Therefore, the context does not support the claim.

grade note: Reverse-inference context lacks ROI-from-same-map circularity; consistent.

D221. NIK-ST-M-003, with-BR, judge no_unrelated [GOOD]

claim: Defining an ROI based on the same statistical mapping used to test functional activation states constitutes circular analysis and yields invalid statistical inference.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The provided reference context discusses factors affecting the BOLD signal and the concept of 'reverse inference' in fMRI. It does not contain any information about defining ROIs based on the same statistical mapping used for testing, circular analysis, or the resulting invalid statistical inference, which are the central points of the claim.

grade note: Context (BOLD factors, reverse inference) off-topic vs circular-analysis/double-dipping claim. Correct.

D222. NIK-ST-M-003, with-BR, judge no_unrelated [GOOD]

claim: Double dipping, using the same data for ROI selection and subsequent selective analysis, results in invalid statistical inference because it violates the criterion that the test statistic must be independent of the selection criteria under the null hypothesis.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The provided reference context discusses factors affecting the BOLD signal and the concept of 'reverse inference' in fMRI research. It does not mention 'double dipping,' ROI selection, invalid statistical inference, or the violation of the independence criterion between test statistics and selection criteria under the null hypothesis. Therefore, the context does not support the specific claim.

grade note: Context lacks double-dipping/independence discussion; BOLD-factors context unrelated. Defensible.

D223. NIK-ST-M-004, with-BR, judge no_unrelated [GOOD]

claim: Collinearity occurs when GLM regressors or linear combinations of regressors are correlated, and near-collinearity is a concern when those correlations are high but not perfect.

source: m4_pubget_papers_bundle_00092.txt

judge reasoning: The claim defines collinearity and near-collinearity in the context of GLM regressors. The provided reference context discusses fMRI preprocessing choices, pipeline effects, and cross-validation models. There is no mention of collinearity, GLM regressors, or related concepts in the context, making the reference entirely unrelated to the claim.

grade note: Preprocessing/CV context has no collinearity/GLM-regressor content; consistent.

D224. NIK-ST-M-005, with-BR, judge no_unrelated [GOOD]

claim: Using extensive nuisance regressors is a common mitigation strategy in task-based fMRI but systematically impacts group-level max t-values and single-subject mean activation.

source: m4_pubget_papers_bundle_00088.txt

judge reasoning: The provided reference context discusses overlap maps and the relationship between group-level and individual-level fMRI effects. However, it does not mention 'nuisance regressors', 't-values', or how any mitigation strategy impacts group-level max t-values or single-subject mean activation. Therefore, the context does not support the specific claim.

grade note: Overlap-map context lacks nuisance regressors/t-values; no_unrelated correctly follows. Both judges agree.

D225. NIK-ST-M-005, with-BR, judge no_unrelated [GOOD]

claim: Systematic comparisons of motion correction strategies in task fMRI show that frame censoring, 6-parameter, 24-parameter, and ICA-based denoising approaches have different efficacy profiles, motivating empirical comparison rather than defaulting to the most aggressive model

source: m4_pubget_papers_bundle_00088.txt

judge reasoning: The claim discusses specific fMRI motion correction strategies (frame censoring, 6-parameter, 24-parameter, ICA-based denoising) and their efficacy. The provided reference context, however, focuses on the utility of overlap maps and the interpretation of group vs. individual effects in multi-subject fMRI studies, particularly concerning compensatory mechanisms in patients. There is no mention of motion correction or any of the specific techniques listed in the claim within the provided text.

grade note: Claim is about motion-correction strategies; cited source is about overlap maps. no_unrelated correct.

D226. NIK-ST-M-006, with-BR, judge no_unrelated [GOOD]

claim: Low cluster-defining thresholds can cause merging of nearby distinct effect clusters, reducing spatial specificity

source: m4_pubget_papers_bundle_00003.txt

judge reasoning: The claim discusses the effect of low cluster-defining thresholds on merging clusters and reducing spatial specificity. The provided reference context discusses nonparametric tests, permutation tests, and randomization tests, as well as experimental design. It does not contain any information about cluster-defining thresholds, their impact on cluster merging, or spatial specificity.

grade note: Permutation-test context lacks cluster-defining-threshold/specificity content; no_unrelated correct.

D227. NIK-ST-M-006, with-BR, judge no_unrelated [GOOD]

claim: Both high and low CDTs create a trade-off between voxel-level false alarm rate and sensitivity to detect widespread effects

source: m4_pubget_papers_bundle_00003.txt

judge reasoning: The claim discusses a trade-off between voxel-level false alarm rate and sensitivity related to high and low CDTs. The provided reference context discusses cluster-based statistics, nominal alpha levels, null hypotheses, and permutation tests, but it does not mention CDTs, false alarm rates, sensitivity, or the specific trade-off described in the claim. Therefore, the context does not support the claim.

grade note: Context is generic cluster stats/permutation; specific CDT false-alarm/sensitivity trade-off absent. Reasonable.

D228. NIK-ST-M-006, with-BR, judge no_unrelated [GOOD]

claim: Low cluster-defining thresholds can merge nearby effect clusters, inflating cluster-extent statistics and increasing false positive risk at the cluster level.

source: m4_pubget_papers_bundle_00003.txt

judge reasoning: The provided reference context discusses general concepts related to cluster-based statistics, nominal alpha levels, null hypotheses, and permutation tests. It does not contain any information about 'low cluster-defining thresholds,' their potential to merge nearby effect clusters, inflate cluster-extent statistics, or increase false positive risk at the cluster level, which are the specific points of the claim.

grade note: Context is generic cluster/permutation stats; specific low-CDT merging/inflation claim absent. Reasonable.

D229. NIK-ST-M-007, with-BR, judge no_unrelated [GOOD]

claim: Temporal filtering should be incorporated into the noise model to best account for changes in degrees of freedom during nuisance regression.

source: m4_pubget_papers_bundle_00010.txt

judge reasoning: The claim discusses the incorporation of temporal filtering into a noise model to account for changes in degrees of freedom during nuisance regression. The provided reference context discusses the importance of understanding analysis choices in fMRI pipelines, focusing on spatial smoothing kernels, HRF modeling, and the inclusion/exclusion of motion regressors as nuisance covariates. It does not mention temporal filtering, noise models, or degrees of freedom, nor does it provide any information that directly or tangentially supports the specific claim.

grade note: Context is smoothing/HRF/motion regressors; claim is temporal filtering/DOF. Specific claim unsupported, verdict fair.

D230. NIK-ST-M-008, with-BR, judge no_unrelated [REVIEW]

claim: Permutation-based neuroimaging inference constructs a null distribution conditional on the observed data, which can support inference at nominal levels but increases computational cost.

source: m4_pubget_papers_bundle_00084.txt

judge reasoning: The reference context mentions 'nonparametric methods for inference such as permutation-based testing may be used' as an alternative to RFT. However, it does not provide any information about permutation-based inference constructing a null distribution conditional on observed data, supporting inference at nominal levels, or increasing computational cost. The context is too generic and does not support the specific details of the claim.

grade note: Context names permutation-based testing as inference alternative, same topic as claim; no_unrelated too harsh, partial fits better.

D231. NIK-ST-M-008, with-BR, judge no_unrelated [GOOD]

claim: Parametric cluster-based statistics, particularly those using low cluster-defining thresholds, have faced scrutiny for false positive (false alarm) rate issues, making exact non-parametric alternatives more valuable.

source: m4_pubget_papers_bundle_00003.txt

judge reasoning: The reference context mentions 'cluster-based statistic' and 'nonparametric tests for neuroimaging data' (specifically permutation tests). However, it does not discuss any scrutiny or issues with false positive rates for parametric cluster-based statistics, nor does it state that non-parametric alternatives are more valuable due to such issues. The context merely introduces non-parametric tests without linking them to the problems described in the claim.

grade note: Context introduces nonparametric/permutation tests but not the false-positive scrutiny driving claim; borderline but defensible.

D232. NIK-ST-M-008, with-BR, judge no_unrelated [REVIEW]

claim: Permutation tests provide exact inference at nominal levels by creating an empirical null distribution conditional on the observed data, which comes at the cost of increased computational effort.

source: m4_pubget_papers_bundle_00084.txt

judge reasoning: The provided reference context only mentions 'permutation-based testing' as a nonparametric method for inference. It does not contain any information supporting the specific claims that permutation tests provide exact inference at nominal levels, create an empirical null distribution conditional on observed data, or come at the cost of increased computational effort.

grade note: Context explicitly names permutation-based testing, the claim's subject; no_unrelated too harsh, should be partial.

D233. NIK-ST-M-008, with-BR, judge no_unrelated [GOOD]

claim: The precision of the empirical null distribution increases with the number of permutations P , but so does computational burden; insufficient P yields imprecise p-values.

source: m4_pubget_papers_bundle_00084.txt

judge reasoning: The reference context mentions 'permutation-based testing' as a nonparametric method. However, it does not discuss the relationship between the number of permutations (P), the precision of the empirical null distribution, computational burden, or the precision of p-values. Therefore, the provided context does not support the specific claim.

grade note: Context only names permutation testing generically; specific P /precision relationship absent. Verdict defensible.

D234. NIK-ST-M-009, with-BR, judge no_unrelated [GOOD]

claim: In a typical fMRI experiment with more than 100,000 voxels, an uncorrected $p < 0.001$ threshold would produce up to roughly 100 false positive voxels, motivating the need for multiple comparison correction.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The provided reference context discusses factors affecting the BOLD signal and the concept of 'reverse inference' in fMRI research. It does not contain any information about the number of voxels, p-value thresholds, false positive voxels, or the need for multiple comparison correction, which are all central to the claim.

grade note: Reverse-inference context lacks voxel-count/p-threshold/false-positive content; no_unrelated correct.

D235. NIK-ST-M-009, with-BR, judge no_unrelated [GOOD]

claim: In standard fMRI experiments, conducting over 100,000 statistical tests per voxel requires robust multiple comparison corrections to avoid unacceptable false-positive rates.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The provided reference context discusses factors influencing the BOLD signal and the concept of 'reverse inference' in fMRI research. It does not contain any information regarding the number of statistical tests per voxel, the necessity of multiple comparison corrections, or the issue of false-positive rates, which are the central elements of the claim.

grade note: BOLD/reverse-inference context lacks any multiple-comparisons content; verdict follows rationale.

D236. NIK-ST-M-009, with-BR, judge no_unrelated [GOOD]

claim: In typical fMRI experiments with 100,000+ voxel-wise tests, uncorrected $p < 0.001$ thresholds can produce up to 100 false-positive voxels, necessitating multiple comparison correction

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The provided reference context discusses factors affecting the BOLD signal and the concept of 'reverse inference' in fMRI research. It does not contain any information about voxel-wise tests, p-value thresholds, false positives, or the necessity of multiple comparison correction, which are all central to the claim.

grade note: Reverse-inference context lacks voxel-test/false-positive content; consistent.

D237. NIK-ST-M-009, with-BR, judge no_unrelated [GOOD]

claim: Low cluster-defining thresholds in cluster-based inference inflate false alarm rates and cause merging of nearby effect clusters, supporting the use of higher cluster-defining thresholds (e.g., $p < 0.001$).

source: m4_pubget_papers_bundle_00003.txt

judge reasoning: The provided reference context discusses cluster-based statistics, nominal alpha levels, and permutation tests in neuroimaging. However, it does not mention 'cluster-defining thresholds,' their impact on 'false alarm rates,' the 'merging of nearby effect clusters,' or the recommendation for 'higher cluster-defining thresholds (e.g., $p < 0.001$).' Therefore, the context does not support the specific claim.

grade note: Context mentions cluster-based stats/permutation but not cluster-defining-threshold effects on false alarms; specific claim unsupported, defensible.

D238. NIK-ST-M-009, with-BR, judge no_unrelated [GOOD]

claim: Low cluster-defining thresholds (CDTs) in cluster-based statistics can cause the merging of nearby effect clusters, heavily reducing spatial specificity.

source: m4_pubget_papers_bundle_00003.txt

judge reasoning: The provided reference context discusses nonparametric tests, permutation tests, and statistical independence, mentioning 'cluster-based statistic' only in a general sense related to alpha levels. It does not contain any information about 'low cluster-defining thresholds (CDTs)', the 'merging of nearby effect clusters', or the 'reduction of spatial specificity' as stated in the claim.

grade note: Context is permutation/nonparametric tests, only generic cluster mention; CDT-merging claim unsupported. Reasonable.

D239. NIK-ST-M-010, with-BR, judge no_unrelated [GOOD]

claim: In standard fMRI experiments with over 100,000 tests, uncorrected thresholds yield unacceptable false-positive rates, requiring robust multiple comparison corrections like FWE or FDR.

source: m4_pubget_papers_bundle_00062.txt

judge reasoning: The provided reference context discusses factors affecting the BOLD signal and the concept of 'reverse inference' in fMRI. It does not contain any information about the number of tests in fMRI experiments, uncorrected thresholds, false-positive rates, or the need for multiple comparison corrections like FWE or FDR, which are the core components of the claim.

grade note: Reverse-inference context lacks FWE/FDR multiple-comparisons content; consistent.

D240. NIK-ST-M-011, with-BR, judge no_unrelated [GOOD]

claim: Very large neuroimaging datasets ($N > 1000$) reveal both statistically significant positive and negative effects across most of the brain, indicating that many directional effects are genuine but that significance alone does not distinguish meaningful from trivial effects.

source: m4_pubget_papers_bundle_00005.txt

judge reasoning: The claim discusses findings from very large neuroimaging datasets ($N > 1000$) regarding statistically significant effects and the distinction between meaningful and trivial effects. The provided reference context describes a methodological procedure involving thresholding steps, LPOs, PPMs, ES thresholds, and ROPE maps for classifying activated/deactivated voxels. While the context touches upon effect sizes and thresholds, it does not mention large datasets, specific findings from such datasets, or the conclusion that significance alone does not distinguish meaningful from trivial effects. The context is purely methodological and does not support the empirical claim.

grade note: Purely methodological threshold context lacks large-N empirical findings; no_unrelated defensible.

D241. NIK-ST-M-011, with-BR, judge no_unrelated [GOOD]

claim: With sufficiently large sample sizes, statistically significant effects are detectable across nearly the entire brain, revealing both positive and negative effects that small studies may miss or misestimate

source: m4_pubget_papers_bundle_00005.txt

judge reasoning: The claim discusses the impact of sufficiently large sample sizes on detecting effects across the brain and avoiding misestimation by small studies. The provided reference context, however, details a methodological procedure involving thresholding steps, LPOs, PPMs, ES thresholds, posterior probability thresholds, HDI, and ROPE maps for classifying voxels. It does not mention sample sizes, their impact on detectability, or the comparison between large and small studies. Therefore, the context is unrelated to the claim.

grade note: Context is voxel-classification thresholding; claim is sample-size effects. Unrelated verdict consistent with rationale.

D242. NIK-TR-H-004, with-BR, judge no_unrelated [GOOD]

claim: Motion-corrupted time points may be removed, but removal has caveats related to temporal contiguity and sufficient remaining time-series length.

source: m4_pubget_papers_bundle_00037.txt

judge reasoning: The provided reference context discusses head motion, motion groups, and temporal drift in fMRI data. It calculates metrics like mean absolute deviation (MAD) and analyzes temporal drift in high movers. However, the context does not mention the removal of motion-corrupted time points, nor does it discuss any caveats related to such removal, specifically temporal contiguity or the length of the remaining time series. Therefore, the context does not support the specific claim.

grade note: Head-motion/drift context never covers scrubbing caveats (contiguity, series length); no_unrelated defensible.

D243. NIK-TR-H-004, with-BR, judge no_unrelated [GOOD]

claim: Motion scrubbing by excising corrupted time-points creates temporal discontinuities that cannot be recombined while preserving temporal contiguity, and centrally located motion-corrupted volumes may render entire segments unsalvageable, introducing artifacts when subsequent interpolation or filtering is applied.

source: m4_pubget_papers_bundle_00037.txt

judge reasoning: The provided reference context discusses temporal drift, motion groups, and sex-based differences in Framewise Displacement (FD). It does not contain any information about motion scrubbing, excising corrupted time-points, temporal discontinuities, unsalvageable segments, or artifacts introduced by interpolation or filtering, which are the central elements of the claim.

grade note: Drift/FD/sex-difference context omits scrubbing discontinuity claims; no_unrelated defensible.

D244. NIK-TR-H-004, with-BR, judge no_unrelated [GOOD]

claim: Motion-corrupted volumes can be excised but require contiguity and minimum-length constraints, otherwise short or fragmented timeseries become unreliable.

source: m4_pubget_papers_bundle_00037.txt

judge reasoning: The provided reference context discusses motion groups, temporal drift in high movers, and sex-based differences in head motion. It does not contain any information about excising motion-corrupted volumes, contiguity or minimum-length constraints, or the unreliability of short/fragmented timeseries, which are the core components of the claim.

grade note: Motion-group/drift context lacks excision contiguity constraints; unrelated defensible.

Judge verdict = cannot_judge

D245. NIK-ST-H-003, with-BR, judge cannot_judge [GOOD]

claim: Treating longitudinal or contemporaneous scans of the same subject as independent data points can inflate classification accuracy from 61% to 86%, with unintentional improvements ranging from 7% to 25% across datasets

source: fpsyt.2019.00983

judge reasoning: The reference is a valid DOI. However, the provided reference context only includes the title, venue, and year of the paper. This metadata does not contain any information about classification accuracy inflation, treating longitudinal scans as independent data points, or specific percentages (61% to 86%, 7% to 25%) mentioned in the claim. Therefore, there is not enough context to determine if the claim is supported by the reference.

grade note: Reference is DOI metadata only (title/venue/year); cannot_judge correctly follows from lack of substantive context.